VTI rapport 1271A

Published 2026

Assessing 15-minute city concept and travel in Sweden

From proximity to practice

Raheb Mirzanamadi and Nina Svensson



vti

vti.se/publications

VTI rapport 1271A

Assessing 15-minute city concepts and travel impacts in Sweden

From proximity to practice

Raheb Mirzanamadi

Nina Svensson



Om inget annat anges är publikationen licensierad enligt CC BY-SA 4.0. Omslaget omfattas inte av licensen.

Unless otherwise stated, the publication is licensed under CC BY-SA 4.0. The cover is not included in the license.

vti

Publikationsuppgifter

Titel

Utvärdering av 15-minutersstaden och dess effekter på resandet i Sverige. Från närhet till praktik.

Författare

Raheb Mirzanamadi (VTI, https://orcid.org/0000-0001-7851-0201)

Nina Svensson (VTI, https://orcid.org/0000-0003-2857-3700)

Utgivare

VTI, Statens väg och transportforskningsinstitut

Serie och nummer

VTI rapport 1271A

Utgivningsår

2026

VTI:s diarienummer

2023/0122-7.2

ISSN

0347–6030

DOI

https://doi.org/10.65151/vti619427

Projektnamn

15-minutersstaden som en möjlighet i Sverige och dess påverkan på resande

Uppdragsgivare

Trafikverket

Publication information

Title

Assessing 15-minute city concepts and travel impacts in Sweden. From proximity to practice.

Author

Raheb Mirzanamadi (VTI, https://orcid.org/0000-0001-7851-0201)

Nina Svensson (VTI, https://orcid.org/0000-0003-2857-3700)

Publisher

VTI (Swedish National Road and Transport Research Institute)

Publication number

VTI rapport 1271A

Published

2026

Registration Number, VTI

2023/0122-7.2

ISSN

0347–6030

DOI

https://doi.org/10.65151/vti619427

Project

The 15-minute city as an opportunity in Sweden and its impact on travel.

Commissioned by

Swedish Transport Administration

Kort sammanfattning

Vårt samhälle är starkt beroende av bilresor, vilket bidrar till miljöproblem, trängsel i trafiken och hälsoproblem. Ett angreppssätt som syftar till att både minska bilberoende och främja aktiv mobilitet är konceptet ”15-minutersstaden”, som prioriterar människocentrerad stadsplanering framför bilorienterad utformning. Denna studie undersöker synen på 15-minutersstaden i två närliggande bostadsområden i Göteborg genom att mäta närheten till viktiga urbana funktioner och analysera invånarnas subjektiva uppfattningar om konceptet.

Studien genomfördes med en metodkombination som omfattade sju djupintervjuer med experter, en enkätundersökning med 216 respondenter och rumsliga kartanalyser. Deltagarna i enkätundersökningen uttryckte blandade uppfattningar om konceptets praktiska effektivitet. Medan många uppskattade den goda tillgängligheten till grundläggande service betonade andra en fortsatt bilberoende vardag för ärenden, kulturella aktiviteter och familjelogistik.

Resultaten visade att även om en hög andel av invånarna (85 % till fots och 100 % med cykel) kan nå grundläggande service inom 15 minuter, kvarstår ett betydande bilberoende, särskilt för arbetsresor, storhandling och besök i köpcentrum. Slutsatserna visar att närhet i sig inte är tillräckligt för att minska bilanvändningen; bekvämlighet, snabbhet och de funktionella fördelarna med privatbilism fortsätter att påverka resebeteenden.

Nyckelord

15-minutersstaden, transporteffektivitet, cykling, gång, hållbar mobilitet, närhet, aktivt resande

Abstract

Our society is heavily reliant on car travel, contributing to environmental degradation, road congestion and various health issues. One approach that aims to simultaneously reduce car dependency and promote active mobility is the concept of the “15-minute city”, which prioritises human-centred urban planning over car-oriented design. This study investigates perceptions of the 15-minute city in two nearby urban areas located adjacent to Gothenburg, Sweden’s second-largest city, by measuring the proximity of key urban functions and examining residents’ subjective perceptions of the 15-minute city concept.

A mixed-methods approach was used to conduct the study, including seven in-depth interviews with experts, a questionnaire survey with 216 respondents, and spatial map analyses. Survey participants expressed mixed views on the concept’s practical effectiveness: while many appreciated the accessibility of essential services, others highlighted continued reliance on cars for errands, cultural activities, and family-related logistics.

The results showed that although a high percentage of residents (85% by walking, 100% by cycling) can access basic services within 15 minutes, car dependency remains significant, particularly for commuting, bulk shopping, and visits to shopping malls. The findings suggest that proximity alone is insufficient to reduce car use; convenience, speed, and the functional advantages of private vehicles continue to shape travel behaviour.

Keywords

15-minute city, transport efficiency, cycling, walking, sustainable mobility, proximity, active travel

Table of contents

Publikationsuppgifter **4**

Publication information **5**

Kort sammanfattning **6**

Abstract **7**

Sammanfattning **9**

Summary **10**

Foreword **11**

1. Introduction **12**

2. Background **14**

3. Method and data **17**

3.1. Interviews 17
3.2. Survey on travel behaviour and perceptions of the 15-minute city 17
3.3. Map analyses 19

4. Results **21**

4.1. Interview findings 21
4.2. Overview of Survey Results 22
4.3. Result of map analysis 30

5. Discussion and policy implementation **34**

6. Conclusion **36**

7. Declaration of generative AI **38**

References **39**

Appendix A. Questionnaire **42**

Appendix B. Comparison of transport mode use between perceived 15-minute city alignment **45**

Area: Partille 45
Area: Mölndal 47

Appendix C. Supplementary Figures **49**

Sammanfattning

Att uppnå klimatneutralitet inom transportsektorn kräver ett grundläggande paradigmskifte, med större fokus på kollektivtrafik och aktiva färdsätt såsom gång och cykel. I detta sammanhang finns ett växande behov av stadsplaneringsstrategier som minskar bilberoendet samtidigt som tillgänglighet och livskvalitet förbättras. Ett sådant tillvägagångssätt är konceptet ”15-minutersstaden”. Detta koncept beskriver urbana miljöer där invånarnas dagliga behov prioriteras framför bilresor och där viktiga tjänster och aktiviteter kan nås på kort tid. I en 15-minutersstad blir gång och cykel de primära färdsätten, vilket gör det möjligt för invånarna att nå de flesta av sina dagliga destinationer inom 15 minuter.

Konceptet 15-minutersstaden är fortfarande under utveckling och en allmänt accepterad definition som kan hantera variationen i urbana miljöer och beteendemönster saknas. Tidigare studier baseras ofta på ett fåtal indikatorer, såsom närhet till grundläggande service. Sådana tillvägagångssätt tar dock inte hänsyn till invånarnas faktiska resvanor – det vill säga om människor utnyttjar service i deras närområde eller fortsätter att förlita sig på bilen trots kort avstånd. Denna kunskapslucka belyser vikten av att identifiera de faktorer som främjar en väl fungerande 15-minutersstad och som kan uppmuntra till kortare resor.

Syftet med denna studie är att undersöka potentialen hos konceptet 15-minutersstaden genom att analysera restiden till olika målpunkter och hur invånarnas resvanor och färdmedelsval relaterar till denna tillgänglighet i två utvalda områden i utkanten av Göteborg. Studien undersöker invånarnas resvanor i de utvalda områdena med avseende på närhet till ett brett spektrum av service och funktioner, inklusive handel, tjänster, skolor, rekreationsområden, kulturinrättningar och vårdcentraler. Analysen bygger på en kombination av intervjuer, enkätdata och spatial analys.

För att få en djupare förståelse om konceptet 15-minutersstaden genomfördes sju djupintervjuer med experter och chefer. Huvudresultaten var att 15-minutersstaden uppfattades som en närhetsbaserad och inkluderande planeringsmodell, men implementeringen möter politiska, institutionella och infrastrukturella utmaningar. Trots att många invånare bor nära grundläggande tjänster är bilanvändningen fortfarande hög.

Enkäten skickades via vykort till 2 000 slumpmässigt utvalda adresser i de valda områdena mellan 22 oktober och 27 november 2024 i Partille och Mölndal. Av de 2 000 utskickade vykorten samlades totalt 216 svar in. Resultaten visade att cirka 75 % av deltagarna uppfattar att deras kvarter delvis eller helt överensstämmer med principerna för en 15-minutersstad. Trots detta använder omkring 82 % fortfarande bilen för större matinköp och skulle fortsätta göra det även om närliggande alternativ fanns, vilket visar att närhet ensam inte ändrar beteendet. Likaså pendlar 40 % med bil – de flesta för resor längre än 15 minuter – främst på grund av snabbhet och bekvämlighet.

Kartanalys genomfördes i de undersökta områdena för att bedöma tillgängligheten för gång och cykel ur perspektivet av en 15-minutersstad. Avstånd mellan bostäder och olika målpunkter beräknades med OSMnx i Python baserat på OpenStreetMaps gatunät. Analysen visade att medan de flesta invånare har god tillgång till grundläggande tjänster, ligger större handelsområden ofta utanför gångavstånd. Även om de går att nå med cykel, gör inköpens karaktär (t.ex. skrymmande varor) att bilen ofta används.

Sammantaget visar resultaten att närhet till service i sig kanske inte är tillräckligt för att minska bilberoendet. När respondenterna exempelvis fick ta ställning till ett scenario där en välsorterad stormarknad låg nära deras bostad uppgav de flesta att de ändå skulle fortsätta använda bil för storhandling. Detta tyder på att närhetens fördelar kan begränsas av praktiska faktorer såsom inköpsvolym, bekvämlighet och behovet av att transportera varor. Vidare innebär inte förbättrad tillgänglighet nödvändigtvis att alla grupper i befolkningen gynnas i samma utsträckning, eftersom olika grupper kan möta olika begränsningar och möjligheter i sina dagliga resor och aktivitetsmönster.

Summary

Achieving climate neutrality in the transport sector requires a fundamental paradigm shift, with greater emphasis on public transport and active modes of travel such as walking and cycling. In this context, there is a growing need for urban planning approaches that reduce car dependency while improving accessibility and quality of life. One such approach is the concept of the "15-minute city." This concept describes urban environments where residents' daily needs are prioritised over car travel, and where essential services and activities can be accessed within a short distance. In a 15-minute city, walking and cycling become the primary modes of transport, enabling residents to reach most of their daily destinations within 15 minutes.

The concept of the 15-minute city is still evolving, with no universally accepted definition or criteria, particularly across different urban contexts and behavioural patterns. Existing studies often rely on a limited number of indicators, such as proximity to basic services. However, such approaches overlook residents' actual travel behaviour—specifically whether people use services situated close to their home or continue to rely on cars despite their proximity. This gap highlights the need to identify the key factors that support a well-functioning 15-minute city and encourage reduced travel distances.

The aim of this study is to examine the potential of the 15-minute city concept by assessing access to key urban services and analysing how residents' travel behaviour and transport choices relate to this accessibility in two selected areas adjacent to Gothenburg, Sweden's second-largest city. The study investigates residents' travel behaviour in the neighborhoods in relation to their proximity to a broad range of amenities, including retail, services, schools, recreational areas, cultural venues, and healthcare facilities. The analysis is based on a combination of interviews, questionnaire data, and spatial analysis.

To gain expert insights into the 15-minute city concept, seven in-depth interviews were conducted with experts and managers. The main findings were that the 15-minute city was seen as a proximity-based and inclusive planning model; however, its implementation faces political, institutional, and infrastructural challenges. Despite many residents living close to basic services, car use remains high.

The survey was distributed via postcards to 2 000 randomly chosen addresses in the selected areas between 22 October and 27 November 2024 in Partille and Mölndal. Out of 2 000 postcards distributed to targeted addresses, a total of 216 survey responses were collected. Survey results showed that about 75% of participants perceive their neighborhoods as aligning with the 15-minute city concept. Nevertheless, around 82% still use cars for bulk grocery shopping and would continue to do so even with nearby alternatives, highlighting that proximity alone does not change behaviour. Similarly, 40% commute by car—most for trips longer than 15 minutes—primarily due to speed and convenience.

Spatial analyses were conducted in the surveyed areas to assess 15-minute city accessibility for walking and cycling. Distances between residences and points of interest were calculated using OSMnx in Python based on OpenStreetMap street networks. Spatial analysis confirmed that while most residents have good access to essential services, larger shopping areas are often beyond walking distance. Although reachable by bicycle, the nature of purchases (e.g., bulky goods) encourages car use.

Overall, the findings indicate that proximity to services alone may not be sufficient to reduce car dependency. For example, when respondents were presented with a scenario in which a large supermarket was located close to their homes, most stated that they would continue using a car for bulk grocery shopping. This suggests that the benefits of proximity may be constrained by practical considerations such as shopping volume, convenience, and the need to transport goods. Furthermore, improving accessibility alone does not necessarily guarantee equal benefits across the population, as different groups may face different constraints and opportunities in their daily travel and activity patterns.

Foreword

This study is financed by the Swedish Transport Administration (grant number TRV 2023/33504) and is conducted at VTI. It is an exploratory research project with the aim to improve the understanding of the 15-minute city in Sweden.

Stockholm, September 2026

Nina Svensson
Project leader

Granskare/Examiner

Chiara Vitrano, VTI.

De slutsatser och rekommendationer som uttrycks är författarens/författarnas egna och speglar inte nödvändigtvis myndigheten VTI:s uppfattning.

The conclusions and recommendations in the report are those of the author(s) and do not necessarily reflect the views of VTI as a government agency.

Publikationen godkänd för publicering/Publication approved for publication

Peter Torstensson, VTI.

1. Introduction

Swedish cities have historically been planned with a strong emphasis on car use (Lundin, 2008), resulting in a high proportion of urban trips being made by car (Trafikanalys, 2024). The infrastructure required for cars—such as roads and parking spaces—occupies a disproportionately large share of urban space. For instance, in Sweden, the average parking space per person is estimated to exceed the average living space—50 m^2 versus 42 m^2 (InfraSweden, 2022). Road traffic also causes other negative impacts, including health effects from noise and air pollution and greenhouse gas emissions (Naturvårdsverket, 2023). In Sweden, in 2024, 35% of greenhouse gas emissions come from road transport, of which 58% are from passenger cars (Naturvårdsverket, 2025). Over the past 15 years, use of biofuels, more energy-efficient vehicles, and electrification have led to reduced CO_2 emissions by about 30% (Naturvårdsverket, 2025; Yang, 2023). However, during the same period, the number of vehicles and total driving distance have increased (SCB, 2023b, 2023a). Even with the ongoing rise in vehicle electrification, problems such as congestion, noise, and particle emissions from tires, road surfaces, and brake wear will persist (Mirzanamadi & Gustafsson, 2022).

The Swedish government aims to reduce CO_2 emissions from road traffic by 70% by 2030 compared to 2010, and to reach zero emissions by 2045 (Ministry of the Environment and Energy, 2018). Achieving climate neutrality requires a paradigm shift in the transport system, focusing on support for public transport and active travel such as cycling and walking (Yannis & Chaziris, 2022). Reducing car travel is an effective strategy for lowering CO_2 emissions and air pollution, while also delivering a range of positive outcomes (Glazener et al., 2022). These include reduced noise, improved traffic safety, increased space for recreation and green areas in urban environments, and decreased resource consumption.

One approach to simultaneously reduce car travel and promote active mobility within an area is the concept of the “15-minute city” (Moreno et al., 2021). The idea of the “15-minute city” was introduced in 2016 by Carlos Moreno, a professor at Sorbonne University (Moreno et al., 2021). The 15-minute city is a concept often used to describe urban environments where people's needs are prioritised over car travel, and where walking and cycling become the primary modes of transport. The name suggests that residents can access their daily needs within 15 minutes by foot or bicycle (Postaria, 2021).

Citizens should easily be able to reach shops, healthcare, schools, pharmacies, recreation, entertainment, libraries, and more, regardless of age, gender, background, or ability (C40 Cities Climate Leadership Group, 2021). For example, car-oriented spaces like street parking can be repurposed to improve walking and cycling conditions. It is also important to emphasise that both physical and digital connectivity are essential to the strategy underlying the 15-minute city. All residents should have access to public transport systems and digital infrastructure, including the internet (C40 Cities Climate Leadership Group, 2021). The concept of the 15-minute city is relatively new. Although various studies (Ferrer-ortiz et al., 2022; Melén & Madsen, 2023; Stockholms Handelskammare, 2023) have examined different sets of indicators—sometimes only a few, such as access to grocery stores and pharmacies—to analyse the concept, there is still no clear definition or specific criteria for what qualifies a city or neighbourhood as a 15-minute city (Knap et al., 2023; Radics et al., 2024), especially when considering a specific city or country where urban characteristics and behavioural norms differ. This lack of consensus highlights the need to explore which factors are essential for a well-functioning 15-minute city and what might motivate people to reduce their travel distances in specific areas. A common approach to analysing the 15-minute city concept in Sweden relies primarily on map-based analyses to assess proximity to urban functions and services (Ferrer-ortiz et al., 2022; Hosford et al., 2022). However, this approach does not account for residents’ travel behaviour, such as whether people actually access nearby services without using a car, or whether they use these services at all, rather than continuing to travel longer distances despite their proximity.

The aim of this study is to investigate the potential of the 15-minute city concept by measuring the proximity of key urban functions and services, and by analysing how residents’ daily behaviours relate to service accessibility and their choice of transport modes, in selected areas in two municipalities—Partille and Mölndal—located adjacent to Gothenburg, Sweden’s second-largest city. The selected area in

Mölndal features a large central shopping centre surrounded by a mix of apartment buildings and single-family houses, along with a concentration of municipal, commercial and public services. The selected area in Partille consists mainly of single-family houses, with some areas characterised by a high concentration of apartment buildings. Services in Partille, including retail, public facilities and local amenities, are more dispersed throughout the municipality, while a major share is also located in the shopping centre just outside its north-eastern boundary.

The study investigates residents' travel behaviour in selected neighbourhoods and analyses their proximity to various categories of amenities (beyond just basic services such as grocery stores, pharmacies, schools, and health centres) including shopping and dining areas, nature and recreational spaces and cultural venues, in order to understand how well the 15-minute city concept aligns with actual accessibility patterns. The study is based on existing literature, interviews, data collection via questionnaires and performing map analysis of two selected areas in the outskirts of Gothenburg.

2. Background

Moreno et al., 2021 introduced the concept of the “15-minute city” based on the idea that urban life improves when less time is spent on transport, especially by car. The concept visualises a city where residents can meet all their daily needs within a 15-minute walk or bike ride, with easy access to six key urban functions: living, working, commerce, healthcare, education, and entertainment. To achieve this, the urban environment should be redesigned according to four core dimensions: density, proximity, diversity, and digitalisation. Density refers to the optimal number of people per square kilometre in an area where urban services function efficiently and resources are used sustainably. Achieving optimal population density enables local authorities to develop tailored solutions for energy production, food supply, and multifunctional use of public spaces. It is important to note that this definition differs from simply maximising the number of residents in ultra-high-rise buildings, which often leads to resource overconsumption, increased energy use, and higher car dependency (Ali Sayigh, 2017; Moreno et al., 2021). Proximity refers to residents’ ability to quickly access essential services—such as shops, schools, and healthcare—within a short distance and time. It enables local residents to make better use of available resources in their neighbouring area for social, economic, and environmental benefits. Diversity refers not only to cultural and demographic variety but also to a balanced mix of residential, commercial, and recreational spaces. This mix ensures that residents can access essential services in their local area without needing to travel far to access them. Digitalisation, provided through internet access and various technologies, refers to services such as online shopping, education, cashless transactions, and virtual communication and working from home. It allows residents to meet their needs without long-distance commuting (Moreno et al., 2021).

It should be noted that the 15-minute time limit is not set in stone. For example, urban planning in America and Australia often rely on 20 minutes, while 15 minutes is more common in Europe and Asia (Staricco, 2022). Also, 30, 10 and 5 minutes are considered in urban planning or accessibility studies (Knap et al., 2023; Staricco, 2022).

Building on this conceptual foundation, various studies have evaluated the concept of 15-minute cities in different urban contexts (Hosford et al., 2022; Melén & Madsen, 2023; Stockholms Handelskammare, 2023). A common limitation in many of these studies is the lack of focus on access to workplaces, even though commuting remains one of the most challenging aspects of urban accessibility (Stockholms Handelskammare, 2023).

Some studies concentrate solely on access to grocery stores. For example, Hosford et al., 2022 studied the concept of the 15-minute city in Vancouver, Canada, by analysing how easily people could reach grocery stores by walking or cycling. The study compared two age groups: under 65 and 65 and older. Results showed that at a walking speed representative of adults younger than 65 years (4.8 km/h), only about 9% of the population lacked access to a grocery store within a 15-minute walk. When a walking speed typical of adults older than 65 years (3.6 km/h) was assumed, accessibility declined substantially: around 21% of the population did not have access to a grocery store within a 15-minute walk, highlighting marked age-related differences in walkable access to grocery stores. Cycling substantially improved grocery store accessibility. At a cycling speed representative of younger travellers (16.2 km/h), residents could reach a large number of grocery stores, more than 10, within 15 minutes, while at a cycling speed typical of older travellers (13.9 km/h), nearly all residents still had access to at least one grocery store within this time threshold. The study also found that areas with limited access to grocery stores had higher proportions of children, older adults, and visible minorities. However, income level and immigration status had no significant impact on accessibility.

A few studies take a broader approach, analysing access to five key urban functions: commerce, healthcare, education, entertainment, and public transport. Ferrer-ortiz et al., 2022 developed a standardised quantitative method to evaluate whether Barcelona, Spain, meets the criteria to be considered a 15-minute city at the micro-scale level. For each key urban function, they defined specific variables. For example, under the healthcare category, the variables included primary care centres, social services, and day centres. In total, 24 sets of variables were established to evaluate people's access to

these services. Accessibility was analysed based on service locations, population distribution, walking speed, and the transport network. Results showed that about 85% of residential areas had access to primary healthcare, education and commerce centres such as supermarkets, markets, fresh food, daily businesses, and catering services. Furthermore, about 95% of residential parcels had access to public transport services and at least one entertainment facility for sports or children. However, access to show centres like cinemas, theatres, and auditoriums was relatively lower, at 72%, due to their concentration in the city's historical centre. Moreover, access to bike lanes in Barcelona was also limited, with only 71% of parcels connected to cycling infrastructure.

Previous studies showed that network-based accessibility analysis is the most commonly used method to assess proximity to key urban functions. For example, Hosford et al., 2022 examined the 15-minute city using city blocks—the smallest geographical units for which population data is available—as origin points for all residents. The Stockholm Chamber of Commerce (Stockholms Handelskammare, 2023) conducted a proximity analysis in Stockholm County using 1,287 existing demographic statistical areas (DeSO), each representing a land segment with approximately 700 to 2,700 residents. In total, 32,565 points of interest corresponding to the location of urban functions were mapped across Stockholm County. The street network was constructed using the Mapbox Isochrone API, which utilises OpenStreetMap data. Ferrer-ortiz et al., 2022 analysed accessibility based on service locations, population distribution, walking speed, and the transport network.

Urban design varies significantly across countries, and different studies emphasise different aspects of urban life. For example, the Stockholm Chamber of Commerce (Stockholms Handelskammare, 2023) found that on average, 26% of Stockholm county's residents live in a 5-minute city and 73% live in a 15-minute city by walking. However, these values vary between the different municipalities within the county. In the City of Stockholm (the most central municipality within Stockholm County), 95% of the population lives in a 15-minute city when walking. Similar patterns were observed in other major Swedish cities (Hatzigeorgiou et al., 2024). In Gothenburg, Sweden's second-largest city, approximately 78% of residents live in areas where a wide range of services, amenities, and activities are accessible within 15 minutes by walking. In Malmö, the third-largest city, accessibility is even higher: 87% of residents live in 15-minute neighbourhoods. However, these studies primarily measured distances to various services, green spaces, and cultural sites and did not analyse in detail how proximity to these amenities affect people's travel behaviour. At the national level, the picture is less optimistic. Only about 30% of Sweden's population lives in areas that meet the criteria of a 15-minute city and in 27 of Sweden's 290 municipalities, the concept is virtually absent—residents in these areas lack access to essential services, amenities, and activities within a 15-minute walk or bike ride (Hatzigeorgiou et al., 2024).

It should be noted that different studies have used varying walking and cycling speeds in their analysis. For the population under 65 years of age, walking speed is typically considered to range between 4.5 km/h and 5 km/h. For individuals aged 65 and older, a lower walking speed of approximately 3.6 km/h is often used. Cycling speeds also vary depending on age, with estimates ranging from 13.9 km/h to 16.2 km/h (Ferrer-ortiz et al., 2022; Hosford et al., 2022; Postaria, 2021).

Traditionally, Sweden has applied various urban models to address the relationship between housing, work, and services—one of the most notable being the ABC city model from the 1950s (Toreld, 2023). This model emphasised proximity between residential areas, workplaces, and commercial centres, much like the 15-minute city. However, it lacked modern priorities such as environmental sustainability, digitalisation and modern work patterns such as working from distance or home. The ABC city model also assumed that people lived near their jobs, which doesn't fit today's lifestyle when several adults in a household may work in different places across the city and many jobs require special skills that aren't available everywhere.

Contemporary planning efforts in Sweden reflect a growing interest in updated principles, such as the project "Street Moves," which aims to take steps toward implementing the 15-minute city concept in Stockholm (ArkDes, 2023). Although the term "15-minute city" is not always explicitly used in planning documents, its core ideas are clearly embedded in current urban development strategies. For example,

“Närastaden” (the Nearby City), a model developed by SWECO, which builds on both the ABC city and 15-minute city concepts (SWECO, 2025), emphasises the importance of adapting infrastructure to support active mobility and integrating digital and ecological considerations into urban design. It reflects a broader shift toward creating more livable, resilient, and sustainable urban environments in Sweden (SWECO, 2025).

Saiduddin’s analysis of urban planning in Sweden’s three largest cities—Stockholm, Gothenburg, and Malmö—shows that all three municipalities are actively working toward goals aligned with the 15-minute city concept (Saiduddin, 2022). These include promoting multifunctional urban areas, expanding walking and cycling networks, and improving access to green infrastructure. Saiduddin’s analysis is aligned with findings from Larsson et al. (2025), who report that the three cities of Stockholm, Gothenburg and Malmö are on average, about 92% accessible by bike and about 86% accessible on foot for destinations related to commerce, education, healthcare, nature and outdoor spaces, and entertainment.

3. Method and data

This section consists of three parts including: (i) interviews, (ii) survey and (iii) map analyses.

3.1. Interviews

To gain insights from experts on the concept of the 15-minute city, we conducted interviews with experts and managers with professional experience in urban planning, regional development and sustainable mobility policy from the Swedish Transport Administration (STA), the municipalities of Gothenburg and Stockholm, Region Västra Götaland and a consulting and research institute in Sweden. Totally seven interviews were conducted based on a semi-structured interview method. In semi-structured interviews, the researcher prepares some questions in advance and focuses on them during the interview. However, the interview remains open, allowing for changes in the order of questions and the emergence of new questions based on the responses given by the interviewees (Patricio et al., 2018).

The interviews lasted approximately 60 minutes and were conducted via online meetings. Each interview was structured around seven main questions, beginning with transport efficiency and subsequently addressing the 15-minute city concept. In this study, transport efficiency is defined following SOU (2022) as a condition in which transport systems are optimised from energy, environmental, and economic perspectives in order to achieve accessibility, sustainability, and competitiveness. This understanding of transport efficiency closely aligns with the principles of the 15-minute city, which emphasise spatial proximity, reduced car dependency, and sustainable mobility choices.

The interview questions are as follows:

(i) How do you define transport efficiency?

(ii) How does your organisation—or your knowledge and policies—address transport efficiency?

(iii) What impact would increased transport efficiency have on your organisation or society?

(iv) Why might the 15-minute city be important?

(v) How can the 15-minute city be promoted?

(vi) What is your organisation doing to support the 15-minute city?

(vii) What are the main obstacles to implementing the 15-minute city?

It is important to note that responses were based on the interviewees' personal experience and expertise. As a result, some topics may not have been covered in full detail or from all possible perspectives. The interviews were conducted in April and June 2024.

The interview material was analysed using a qualitative thematic analysis. All interviews were transcribed and coded to identify common themes related to transport efficiency and the 15-minute city. The coding process began by identifying common topics in the interviews, which were then grouped into broader themes to compare similarities and differences across interviewees. Themes were refined through repeated readings of the transcripts, capturing both predefined questions and new issues raised by the interviewees.

3.2. Survey on travel behaviour and perceptions of the 15-minute city

Based on insights from previous studies and interviews, a web-based survey was designed to collect data on individual mobility patterns and perceptions of the 15-minute city. The survey included 31 questions. The questionnaire is shown in Appendix A. Participants were asked to reflect on their travel habits and activities over the past month. The questions covered a range of topics including working status, transport mode usage, commuting time, number of days working from home, reasons for choosing different

transport modes, acceptable cycling durations, and barriers to walking or cycling. Respondents were also asked which transport modes they use for various activities, whether certain destinations require travel beyond their local area, and whether large and well sorted grocery stores are accessible nearby. To assess perceptions of the 15-minute city, participants were asked whether they consider their local area to meet the criteria of a 15-minute city, and if not, what key elements are missing. They were also asked whether the concept could work well for them personally and how it might influence their choice of transport mode. The survey concluded with general background questions, including gender, income, education level, number of children, and housing type.

The survey was prepared in both Swedish and English and distributed in two selected areas in Gothenburg. The two areas shown in *Figure 1* were selected based on several criteria. First, they had to be located on the outskirts of a large city and be more than a 15-minute bike ride from the city centre, ensuring that they were clearly separated from central Gothenburg. Second, they needed to be well connected to Gothenburg by public transport, so that most residents would not require a car for commuting. Third, the areas had to contain a mix of housing types and some commercial functions, giving them the potential to operate as a 15-minute city.

By using these criteria, two areas emerged, one east and one south of Gothenburg city centre, in the municipalities of Partille and Mölndal respectively. Partille has a population of approximately 41 thousand residents, Mölndal 71 thousand, and Gothenburg 610 thousand (Göteborg stad, 2024; Mölndal kommun, 2025; Partille Kommun, 2025). After choosing the general location of the areas, a number of postal-code areas were chosen. The selection started from the centee of the chosen areas and postal-code areas were added until there was a mix of approximately 50 % single-family houses and 50 % apartments and at least 1000 addresses. The drawback of such an approach is of course that there may be some unknown patterns in the chosen areas that differ from the surrounding areas. However, this would be hard to identify beforehand. In Partille, eight postal zones containing 1,310 individual addresses were selected, and in Mölndal, eleven postal zones with 2,228 addresses were selected. Postal-code areas were used because it is the finest geographical unit at which it is possible to request addresses from the Swedish state personal address register and because it is less intrusive to ask respondents for their postal code in the survey rather than their full address.

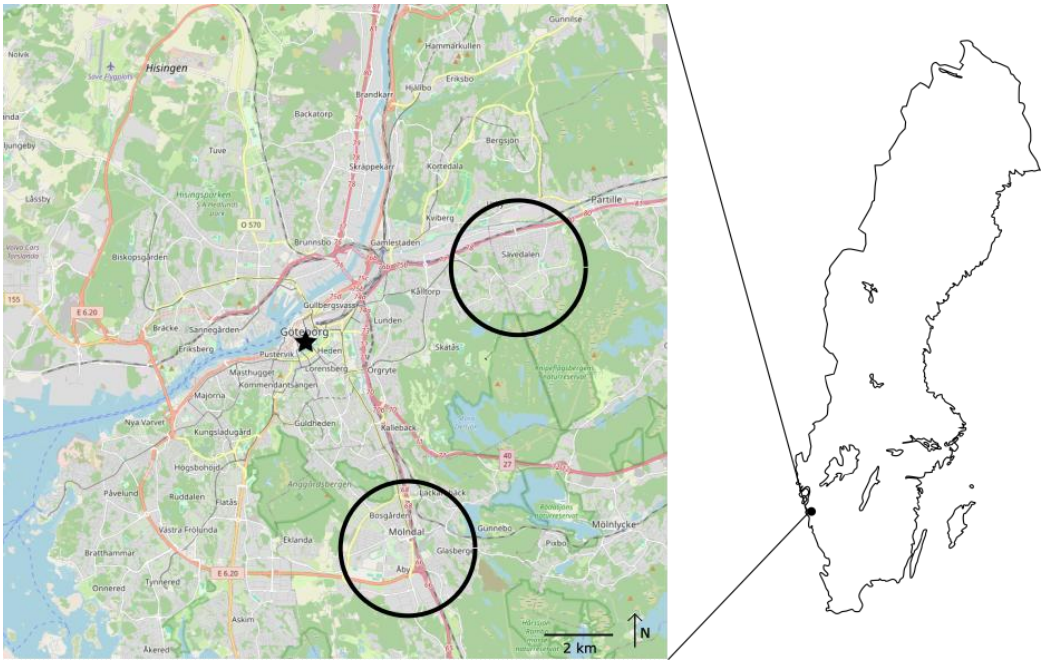


Figure 1. Map of Sweden and Gothenburg, with the two selected areas marked by cirles. The star marks Gothenburg city centre. Map source: OpenStreetMap

The survey was distributed via postcards to 2000 randomly chosen addresses in the selected areas between 22 October and 27 November 2024. This time window was selected to avoid the autumn school holidays in Sweden as well as potential disruptions caused by early winter weather.

The target group included residents living in the selected addresses, aged 18 and older. It should be noted that the survey was sent to a specific individual, not to a household. This means that several people in the same household might receive the survey.

The survey responses were analysed using descriptive statistical methods. Frequencies and percentages were calculated to examine respondents' socio-demographic characteristics, commuting patterns, and perceptions related to the 15-minute city. The results were summarised using tables and graphical visualisations to illustrate modal share, commuting time, reasons for mode choice, perceived barriers to active travel, and mode use across different types of daily activities. Furthermore, postal code–level data were used to compare the municipalities of Mölndal and Partille in order to identify similarities and differences.

3.3. Map analyses

Map-based spatial analyses were conducted in the selected areas to evaluate whether they meet the criteria of a 15-minute city, specifically in terms of walking and cycling accessibility. The map analyses were performed in the smallest rectangular areas that could fit all the postal code areas. However, the results focus on the areas within the chosen postal codes.

Four key urban functions were examined: commerce, healthcare, education, and entertainment. Due to limitations in data availability and the variability of work arrangements (e.g., working from home vs. commuting), workplace accessibility was not included in the map analysis. Instead, data related to commuting and work-from-home frequency were collected through the survey component of the study.

For the commerce category, points of interest included grocery stores suitable for weekly shopping (excluding very small shops with limited selection), shopping areas with a variety of stores (e.g., clothing, furniture), and places to eat (including fast food, casual dining, and fine restaurants, but excluding cafés). For the grocery store inclusion criteria, the purpose was to choose only stores with large enough selection that a normal person's needs would be satisfied without having to visit several stores. Relatively small (such as ICA nära), medium-sized stores and hypermarkets were included. The reason to exclude cafés from places to eat was to distinguish between different categories, i.e. places that you would visit to eat a full meal versus places you would go to for leisure. While both types of places are relevant, establishments offering full meals were considered to more directly fulfil essential daily needs.

Healthcare facilities were defined as pharmacies and health centres. Education included daycare and schools offering grades 0–3 and 4–6; schools offering up to grade 5 were also included. Entertainment encompassed natural areas, cultural venues, and spaces for social activities. Natural areas were included only if they measured at least 200 metrer along their longest axis or were clearly designated for recreation, such as parks. Cultural and social venues included publicly accessible places such as cinemas, theatres, libraries, museums, adventure parks, and sports facilities. Only venues and activities that are open to everyone—regardless of religion, political affiliation, or nationality—were included in this category. This gives in total ten categories. All points of interest were identified using a combination of Google Maps and targeted web searches. Points within and up to 1.5 km outside the rectangular map study area were included.

Distances to each point of interest were calculated using the OSMnx module in Python, which processes map data from OpenStreetMap. A street network for walking and biking was extracted from OpenStreetMap. This includes both roads for motor traffic with lower speed limit and separated walking or cycling paths. In the street network every intersection and turn were represented as a node. These nodes were used to approximate residential addresses. Although these nodes do not correspond exactly to residential addresses, they provide a reliable approximation due to the structure of urban street networks. Using OSMnx, the shortest path from a point of interest to every node in the network was

calculated. This process was repeated for all points of interest in a category, including those located just outside the study area boundaries. This produced ten maps for each selected area, representing the distance from each node to the nearest point within that category. The ten maps were then combined into a single map, which shows the distance to all categories. When combining the maps, the maximum value in every node was chosen. In this way, all categories were made equally important. For example, if the distance from a node to the closest grocery store was 100 m, the distance to the closest pharmacy was 1 km and the distance to the closest shopping area was 2 km, that node will show a distance of 2 km. It thus means that all three categories can be found within 2 km from this node. In practice, this means that the combined map will resemble the map with the most sparsely distributed category. After this, the distances in the combined map were interpolated into a continuous grid using cubic spline interpolation, enabling clearer visual representation and comparative analysis of accessibility across the study areas. Finally, the map was recalculated into walking and cycling time. In this study, walking speed was fixed at 5 km/h and cycling speed at 15 km/h, independent of topographical conditions.

4. Results

This section included three sections: (i) interview findings (ii) overview of survey results and (iii) results of map analyses.

4.1. Interview findings

Interviewees referred to transport efficiency as a holistic approach to mobility that goes beyond optimising transport systems or reducing traffic volumes, encompassing reduced resource use, improved accessibility, and broader societal benefits. They connected it to urban planning goals such as reducing motorised traffic, encouraging walking and cycling, and enhancing public health. Additionally, they emphasised the influence of behavioural and psychological factors—such as social norms, individual attitudes, and the quality of the travel environment—on transport choices.

All interviewees described practical efforts to promote transport efficiency, including its integration into strategic documents and infrastructure measures like reallocating street space, expanding pedestrian and cycling networks, and investing in public transport. They also highlighted regulatory and behavioural tools, such as parking policies and speed limits, aimed at influencing travel behaviour, reducing car dependency, and supporting the transition to more sustainable transport modes. Interviewees identified potential impacts, including shifts in organisational responsibilities, reduced need for large-scale infrastructure, environmental and health improvements—such as better air quality, lower noise levels, and increased physical activity—enhanced urban quality, reduced societal and individual costs, alignment with sustainable development goals, and recognition of political and institutional barriers to traffic reduction.

The concept of the 15-minute city was addressed in all seven interviews, though the degree of engagement with the term varied. Interviewees described the 15-minute city as a desirable and inclusive planning principle centred on spatial proximity, walkability, access to everyday services, and vibrant public spaces. They viewed it as both a modern evolution of earlier planning models and a foundation for creating compact, cohesive urban environments. They revealed both strong support for the concept and significant structural, political, and spatial challenges to its realisation.

Two interviewees expressed that many Swedish cities already meet the criteria of a 15-minute city, depending on how the concept is defined, and a large share of the population lives within short distances to basic services. However, both also noted that despite this proximity, car dependency remains high, partly due to planning practices that prioritise car access.

Regarding how the 15-minute city can be promoted, the interviewees described various strategies. These included improving walking and cycling infrastructure, integrating services into residential areas, reducing traffic barriers, and encouraging mixed-use development.

They emphasised the importance of collaboration—both within municipalities and with external actors such as researchers, civil society, and the private sector. Several interviewees noted that such collaboration is necessary to align land-use planning, transport provision, and service location. One interviewee highlighted the need for long-term planning consistency, noting that proximity-based development requires sustained political commitment and continuity over time. Meanwhile, another pointed to the role of behavioural change and public engagement. emphasising that spatial proximity alone is insufficient if residents do not perceive walking, cycling, and public transport as safe, attractive, and socially supported options.

In terms of concrete actions, five interviewees described ongoing efforts within their organisations aligned with the 15-minute city concept. These included consequence analyses and accessibility studies to evaluate proximity to essential services, as well as mobility measures such as establishing local service hubs, creating neighbourhood workspaces, and enhancing infrastructure to support walking, cycling, and public transport. Public communication, awareness-raising initiatives, and engagement with schools, workplaces, and local communities were also described as important complements to spatial and

infrastructural measures. All seven interviewees identified various obstacles to implementing the 15-minute city. Commonly mentioned barriers included traffic and infrastructure-related obstacles, such as large roads, railways, and geographic features like water bodies. They pointed to political or institutional challenges, including the lack of a shared goal across municipalities, political polarisation around the term, and planning traditions that favour regional enlargement and car-oriented development. One interviewee noted that while some municipalities use terms like "compact and cohesive city," these may still prioritise speed and distance over true proximity.

4.2. Overview of Survey Results

Out of 2000 postcards distributed to targeted addresses, a total of 216 survey responses were collected. *Table 1* provides an overview of the demographic characteristics of the respondents, segmented by the municipalities of Mölndal and Partille, as well as the total sample. According to SCB (Statistics Sweden, 2025), the average age in these municipalities is approximately 40 years, the gender distribution is nearly equal, and around 83% of individuals aged 20 to 64 are employed. A comparison between the survey data from *Table 1* and official statistics reveals that older individuals and retirees were overrepresented in the survey. However, given the study's focus on the 15-minute city concept —regardless of commuting to work— the collected data remains valuable for analysis, particularly in understanding accessibility and mobility patterns among different demographic groups.

Table 1. Sample Characteristics

Category	Variable / Statistic	Mölndal	Partille	Total
Employment status (number of responses)	Employed	63	69	132
	Student	6	1	7
	Unemployed or retired	38	39	77
Gender (%)	Female	45.7%	47.8%	46.8%
Age (years)	Mean (SD)	53.2 (18.1)	56.9 (15.6)	55.1 (16.8)
Household composition	Number of people in household, mean	2.3	2.7	2.5
	Have children (%)	49.5%	56.9%	53.2%
	Number of children, mean	0.4	0.8	0.61
Work from home	Days WFH per week, mean (SD)	0.9 (1.3)	1 (1.3)	0.9 (1.3)
Education level (%)	Primary school	3.7%	2.8%	3.2%
	High school	20.6%	15.9%	18.1%
	Post-secondary education	9.3%	15.9%	12.5%
	College/university degree	60.7%	59.6%	60.2%
	PhD degree	5.6%	6.5%	6.0%
Income	Average household income (SEK), mean	70 000 SEK	75 143 SEK	72 595 SEK
	0- 30 000 SEK (%)	12.1%	10.1%	11.1%
	30 000 – 60 000 SEK (%)	29.0%	28.4%	28.7%
	60 000- 90 000 SEK (%)	23.4%	22.9%	23.1%
	90 000- 120 000 SEK (%)	20.6%	16.5%	18.5%
	More than 120 000 SEK (%)	7.5%	18.3%	13.0%
	Prefer not to say (%)	7.5%	3.7%	5.6%
Housing type (%)	Apartment	64.5%	58.7%	61.6%
	House/row house	35.5%	41.3%	38.4%

One factor in spatial analyses related to the 15-minute city is commuting to workplaces, although such analyses are often constrained by the difficulty of accessing reliable data on workplace locations. To address this limitation, this study included questions in survey asking participants to report their typical mode of commuting between home and work or study (i.e., the one they spend the most time using). *Figure 2*(a) presents the modal share for commuting, showing that the car is the dominant mode at approximately 40%, followed by cycling and public transport. *Figure 2*(b) illustrates commuting times, revealing that only 23.9% of participants reach their workplace or study location within 15 minutes, while 71.6% do so within 30 minutes. Furthermore, *Figure 2*(c), shows that over 80% of participants who commute primarily by car reach their workplace or study location in less than 30 minutes. Among public transport users, more than 90% have a commute time exceeding 15 minutes. For cyclists, over 75%

reported a travel time of less than 30 minutes. In the case of walking, more than 70% of participants reach their workplace in under 15 minutes, and all walking commuters arrive within 30 minutes. These findings suggest that work-related travel remains an important component of daily mobility that is not fully captured by residential accessibility analyses. As workplace locations were not explicitly included in the spatial analysis, the results should be interpreted with caution.

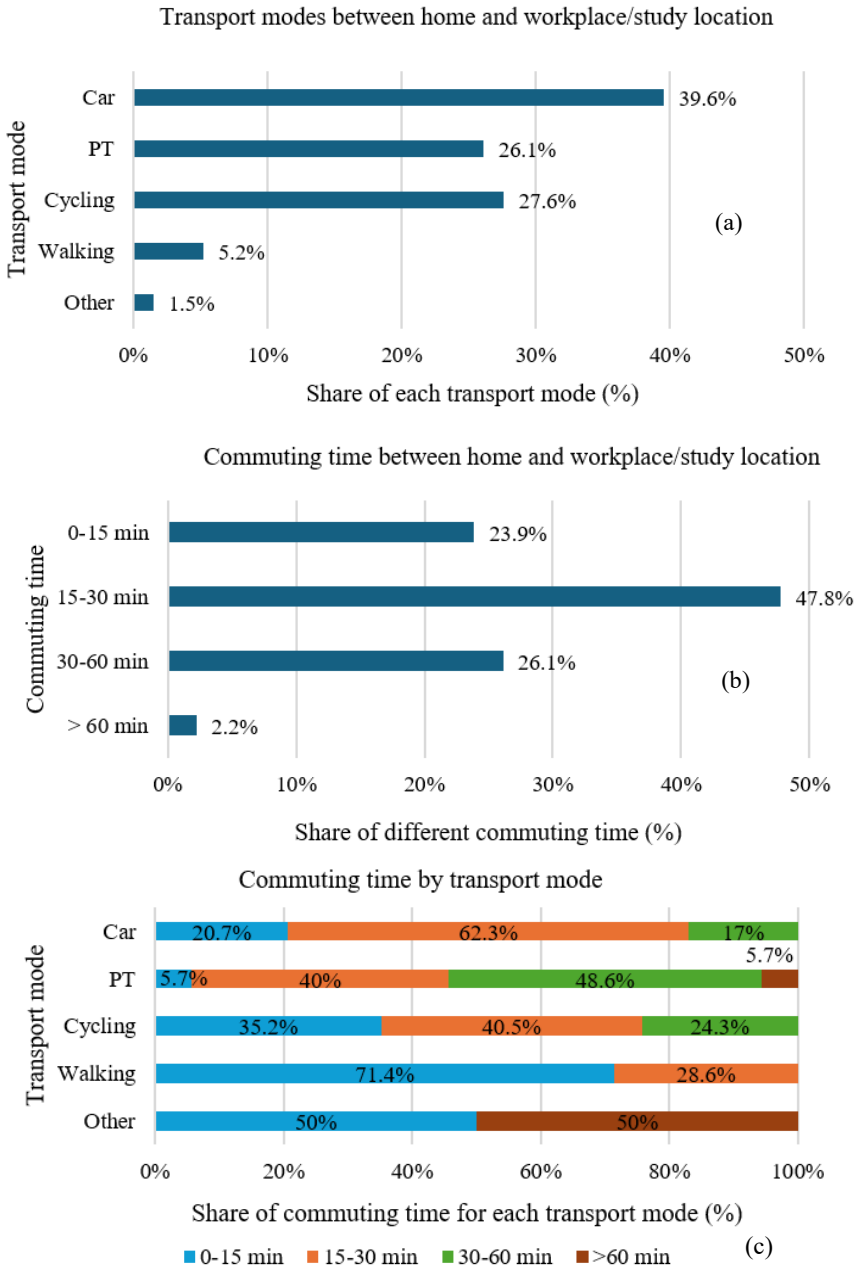


Figure 2. Modal share and time for commuting *between home and workplace/study location (a) modal share (b) travel time in one direction (c) commuting time by transport mode (Total number of responses were 134, all of whom worked or studied and worked from home a maximum of 3 days per week).*

To better understand the reasons behind participants' choice of commuting modes—car, public transport, cycling, and walking—a set of factors were investigated for each mode. These included convenience, personal preference, affordability, speed, lack of viable alternatives, health benefits, environmental considerations and other unspecified reasons. In addition, functional suitability, such as the need for specific vehicle features (e.g., cargo space or capacity for multiple passengers), was included for car use. *Figure 3* presents the distribution of these factors across different transport modes, used for commuting to work. For car users, the most commonly cited reason was speed, with 41% of

participants indicating that driving is the fastest way to reach their workplace, followed by convenience (32%). For public transport, affordability and environmental friendliness were the most frequently mentioned reasons (both at 20%), followed by convenience (17%). In the case of cycling and walking, the dominant factors were cost-effectiveness, personal preference, and health benefits, each cited by over 50% of respondents. Additionally, speed was noted as an important factor for cycling, while 16% of walking commuters also mentioned travel time as a relevant consideration. Compared to other transport modes, few people considered public transport to be a fast mode for commuting to work. This is understandable, as both Mölndal (approximately 7 km from central Gothenburg) and Partille (around 10 km from central Gothenburg) are located at a moderate distance from the city, which is likely the primary workplace destination for many participants. Public transport in these areas may involve multiple stops and, in some cases, transfers. It is hypothesised that when considering door-to-door travel, this may contribute to longer waiting and travel times compared to more direct modes such as driving or cycling, particularly for trips not oriented toward the main city centre.

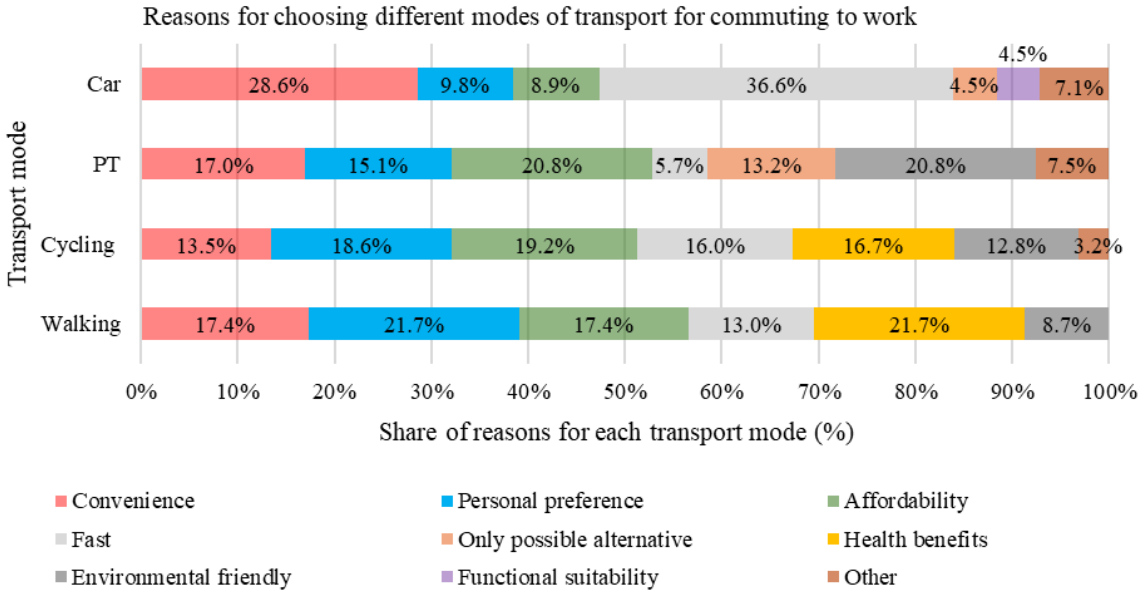


Figure 3. Share of participants' reasons *for choosing different modes of transport for commuting to work.*

Although car use remains the dominant mode, active modes—cycling and walking—play a central role in the 15-minute city concept and are therefore examined in greater detail in the continuation of this study. To assess the maximum acceptable cycling/walking time, all survey respondents were asked to indicate their preferred upper time limit for running errands (e.g., shopping, picking up packages, or dropping off and collecting children). In addition, participants were asked to identify common barriers to cycling/walking in their local area.

Of the 216 survey responses, 13% indicated a maximum acceptable cycling/walking time of 5 minutes, 63% up to 15 minutes, 20% up to 30 minutes, and only 4% were willing to cycle/walk for more than 30 minutes to run errands. These results suggest that most participants are open to cycling/walking for short (up to 15 minutes) to moderate durations (up to 30 minutes), which has implications for planning local infrastructure and services within cycling/walking-friendly distances.

Additionally, participants were asked to identify common barriers to cycling/walking, with the option to select multiple responses. In total, 234 barrier responses were collected. When calculated based on the number of responses, 47% reported no significant hindrance, while others highlighted several challenges. As shown in *Figure 4*, the most frequently cited barriers included lack of bicycle repair facilities and other unspecified reasons. Further responses revealed a range of barriers that discourage cycling/walking. Physical challenges such as hilly terrain, strenuous effort, and concerns about sweating, particularly before work or appointments, were frequently mentioned barriers for cycling. Health-related limitations,

including chronic conditions, age-related impairments, and reduced mobility, also played a significant role. Adverse weather conditions like rain, snow, and ice were cited as deterrents, along with infrastructure issues such as poor road maintenance, frequent construction, and a lack of safe, direct cycling paths. Practical concerns, including difficulty transporting items, fear of theft, and lack of access to a bicycle, were also noted. Finally, personal preferences, discomfort, and negative experiences with other cyclists contributed to the reluctance to cycle.

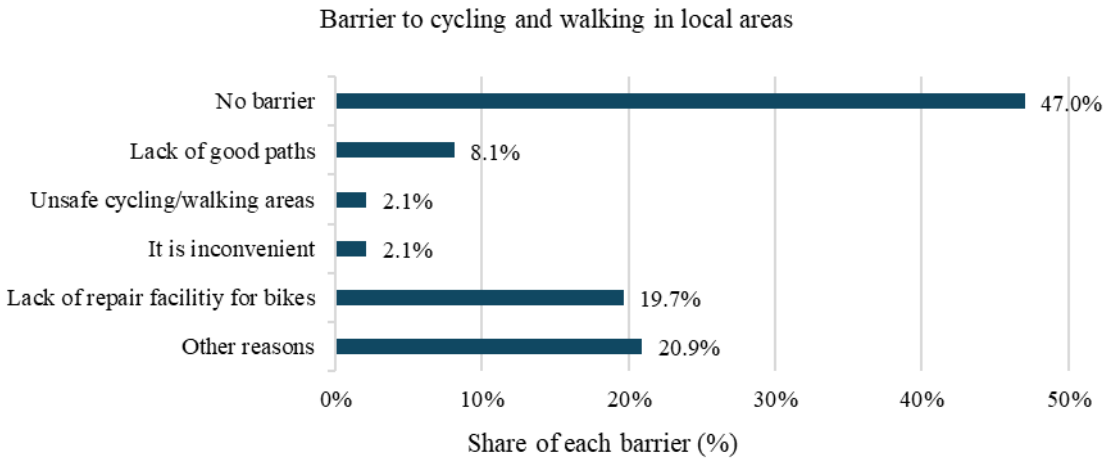


Figure 4. Percent of responses for each reported barrier to cycling and walking among participants (In total, 234 responses were collected, as participants were allowed to select multiple options).

To understand how different transport modes are used for various daily activities, participants were asked to indicate which modes they typically use for specific purposes. The survey allowed multiple responses per participant. The listed activities included: bulk grocery shopping (e.g., buying for a whole week), regular grocery shopping, running errands (e.g., collecting packages, visiting healthcare centres), physical activity (e.g., gym, outdoor training), visiting restaurants and cafés, cultural activities (e.g., cinema, theatre, library), leisure activities, and escorting children to school or kindergarten.

Figure 5 presents the distribution of transport modes by activity type, illustrating which modes are typically used for different everyday activities. In this analysis, the category “Other/Not doing this activity” was excluded, as it combines respondents who use alternative modes with those who do not perform the activity, which may lead to misinterpretation of transport mode choice. Therefore, only Car, Public Transport, and Bike/Walk were considered.

For bulk grocery shopping and leisure activities, the car is the most frequently used mode. Public transport is most commonly used for visiting restaurants and cultural venues. This pattern likely reflects the spatial concentration of such destinations in central Gothenburg, where parking is often limited and expensive, making public transport a more practical option. Cycling and walking are primarily used for regular grocery shopping (excluding bulk shopping), running errands and accessing training facilities.

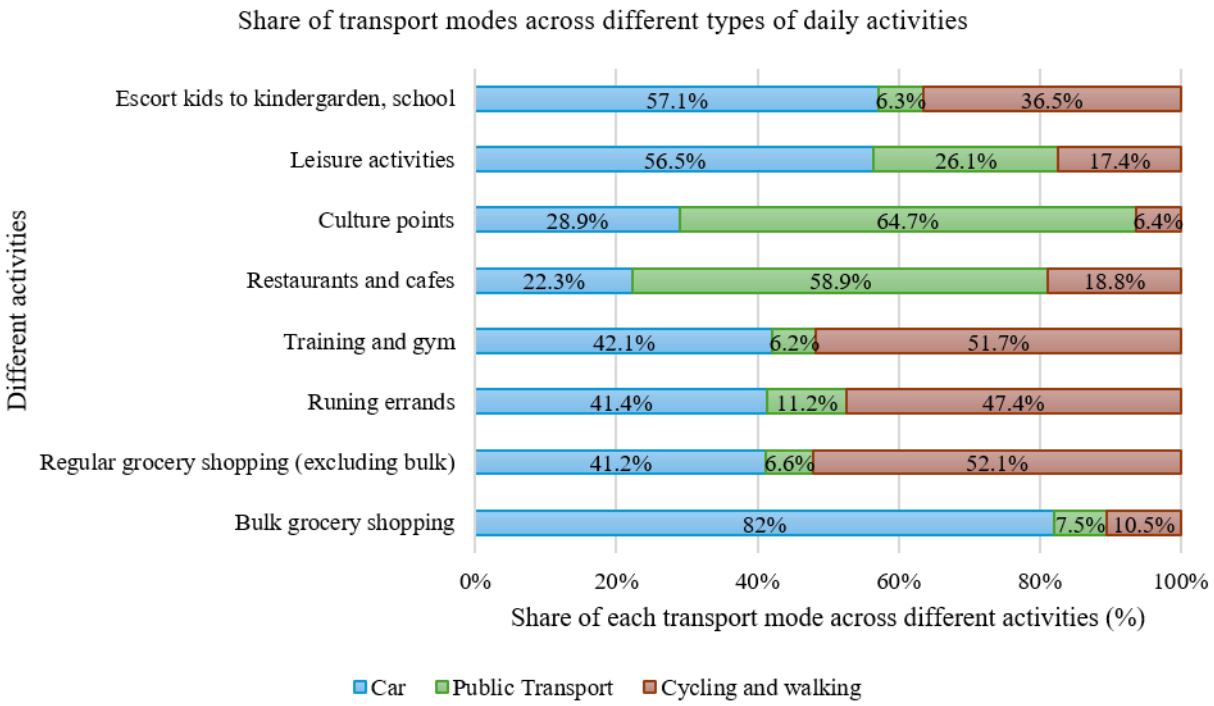


Figure 5. Share of transport modes across different types of daily activities (Total collected responses for each activity was as following: 200 for bulk grocery shopping, 211 for regular grocery shopping (excluding bulk), 215 for running errands, 178 for training and gym activities, 202 for visiting restaurants and cafés, 204 for cultural activities, 207 for leisure activities, and 63 for escorting children to kindergarten or school).

As shown in *Figure 5*, 82% of bulk grocery shopping trips were reported to be made by car. Given the total collected responses for bulk grocery shopping excluding "Other/Not doing this activity" is 200, this corresponds to 164 responses. These participants were subsequently asked whether they would consider alternative transport modes if a well-stocked, large supermarket was located near their homes. Among them, 65% stated that they would still use a car. Meanwhile 16% indicated they would switch to cycling or walking, and only 5% reported they would use public transport. Additionally, 14% were unsure. These results suggest that proximity to services alone may not be sufficient to shift travel behaviour. One possible explanation is that bulk or weekly shopping is inherently associated with larger purchase volumes, which may limit the practicality of active or public transport modes, regardless of distance. It is also plausible that increased proximity could encourage a shift away from bulk weekly shopping toward more frequent, smaller trips conducted by active transport. As such, the results should be interpreted with caution, as they reflect not only the role of proximity but also the complex relationship between distance, shopping frequency, and shopping volume.

Another interesting question related to the 15-minute city concept concerns why people travel long distances for certain activities. To explore this, participants were asked to identify which services they travel far to access because they are lacking or of low quality in their local area, with the option to select multiple responses. The alternatives included: larger and better-stocked grocery stores, improved schools and preschools, enhanced training facilities (e.g., gyms, sports centres, outdoor exercise areas), more and higher-quality restaurants and cafés, richer social and cultural activities, and larger or more accessible natural areas such as parks, forests, and beaches.

Figure 6 illustrates the proportion of respondents who travel long distances to access different activities. Percentages are calculated as the share of respondents (N = 216) selecting each activity, with multiple responses permitted. The most frequently cited activities or destinations for which respondents travel long-distances were to access to more/better restaurants, larger and more well-stocked grocery stores, more/better cultural venues, and more/larger natural areas.

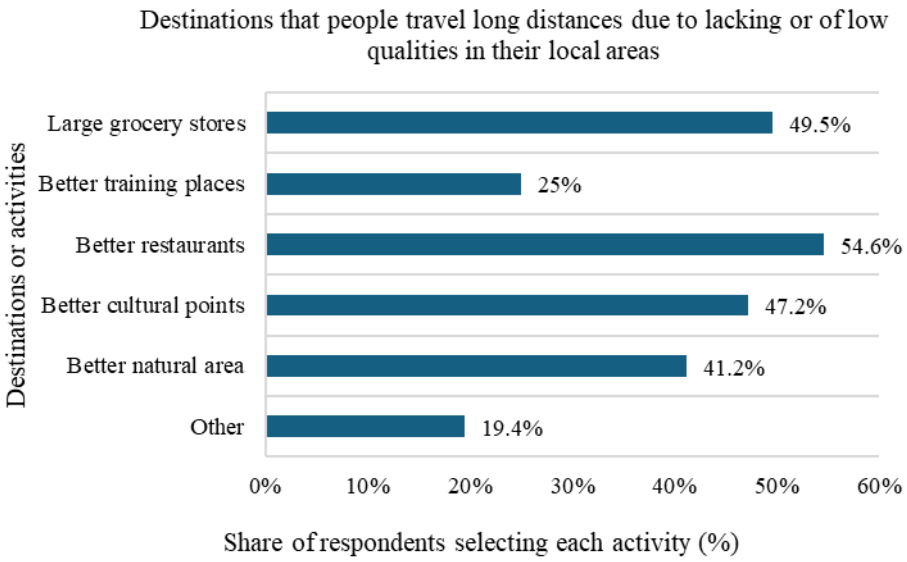


Figure 6. Share of respondents who travel long distances to access different activities in response to missing or of low-quality local services.

In addition to these service-related gaps, responses to the open-ended "Other" option revealed a broader range of motivations for traveling beyond the immediate residential area. Social and family connections play a major role, as individuals often visit relatives or attend gatherings in other parts of Sweden. Recreational and leisure activities also contribute significantly, including trips to summer houses, beaches, golf courses, or participation in organised sports and games such as chess. Access to shopping and services remains a key factor, with many seeking larger retail centres or specialised stores not available locally. Educational pursuits, such as adult evening courses or swimming lessons, further encourage mobility. Religious and community engagement, particularly when local options are limited or inaccessible by public transport, also prompts travel. Employment-related reasons, including commuting to workplaces or visiting job-seeking centres, are common. Lastly, while some residents report having sufficient amenities nearby, others travel to access better healthcare, facilities, or services that are not available in their immediate vicinity.

Participants were also asked whether they considered their neighbourhood aligning with the principles of a 15-minute city—defined as an area where all daily needs are accessible within 15 minutes by walking or cycling, supported by safe pedestrian and cycling infrastructure and frequent public transport. The results are presented in *Figure 7*. As shown, 84% of participants living in the selected areas in Mölndal municipality reported that they considered their area to fully or partially meet the criteria of a 15-minute city. In contrast, only 63% of participants from the selected areas in Partille municipality shared this view. *Figure 8* (left) presents participants' responses by postal code in Mölndal, while *Figure 8* (right) displays the corresponding responses for Partille. The results by postal code are particularly interesting in relation to the map analysis in Section 4.3. They show that when people live within walking distance of essential services, amenities, and activities, their perception of their area as a 15-minute city is higher compared to those who live farther away from such resources. While this result follows intuitively from the core definition of the 15-minute city, it is nevertheless important because it empirically demonstrates that residents' subjective perceptions align closely with objectively measured spatial accessibility. In other words, the concept of the 15-minute city is not only theoretically defined by proximity, but is also meaningfully reflected in how residents experience and evaluate their local environment.

A comparison of car use and cycling between individuals who perceived their area as fully or partially aligned with the 15-minute city concept (Group 1) and those who did not or were unsure (Group 2) revealed differences in travel behaviour. In this analysis, the category "Other/Not doing this activity" was excluded from the analysis, as it combines respondents who use alternative modes with those who do not perform the activity. This issue is particularly relevant for activities such as escorting children, where a

large share of respondents may not be involved. Therefore, only Car, Public Transport, and Bike/Walk were considered. In the Mölndal area, Group 1 showed, on average, a 17.9 percentage point lower share of car use and a 17 percentage point higher share of cycling or walking compared to Group 2. In Partille, the same trend was observed: Group 1 used cars 2.2 percentage point less and cycled or walked 6.6 percentage point more than Group 2. The results of this analysis are presented in Appendix B. The distribution of car use and walking or cycling across different daily activities for both municipalities is presented in *Figure 9*.

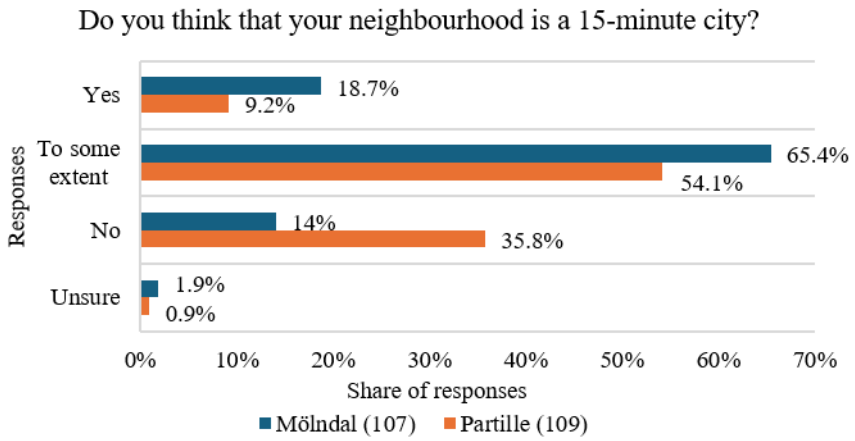


Figure 7. Share of responses reflecting participants' perceptions *of whether their neighborhood aligns with the principles of a 15-minute city*

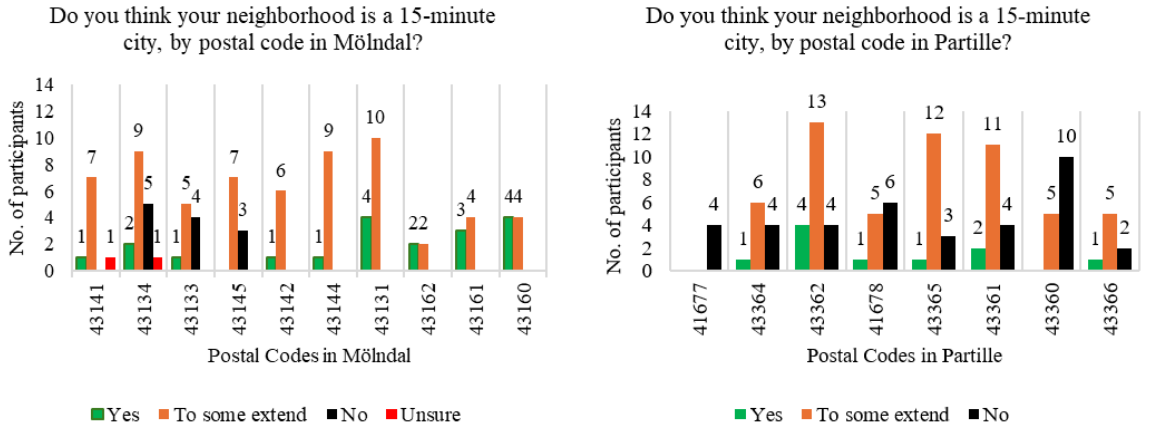


Figure 8. Participants' perceptions *of whether their neighbourhood aligns with the principles of a 15-minute city. (Left) number of responses by postal codes in Mölndal, (right) number of responses by postal codes in Partille (Postal codes can be seen in Figure 10 and Figure 11).*

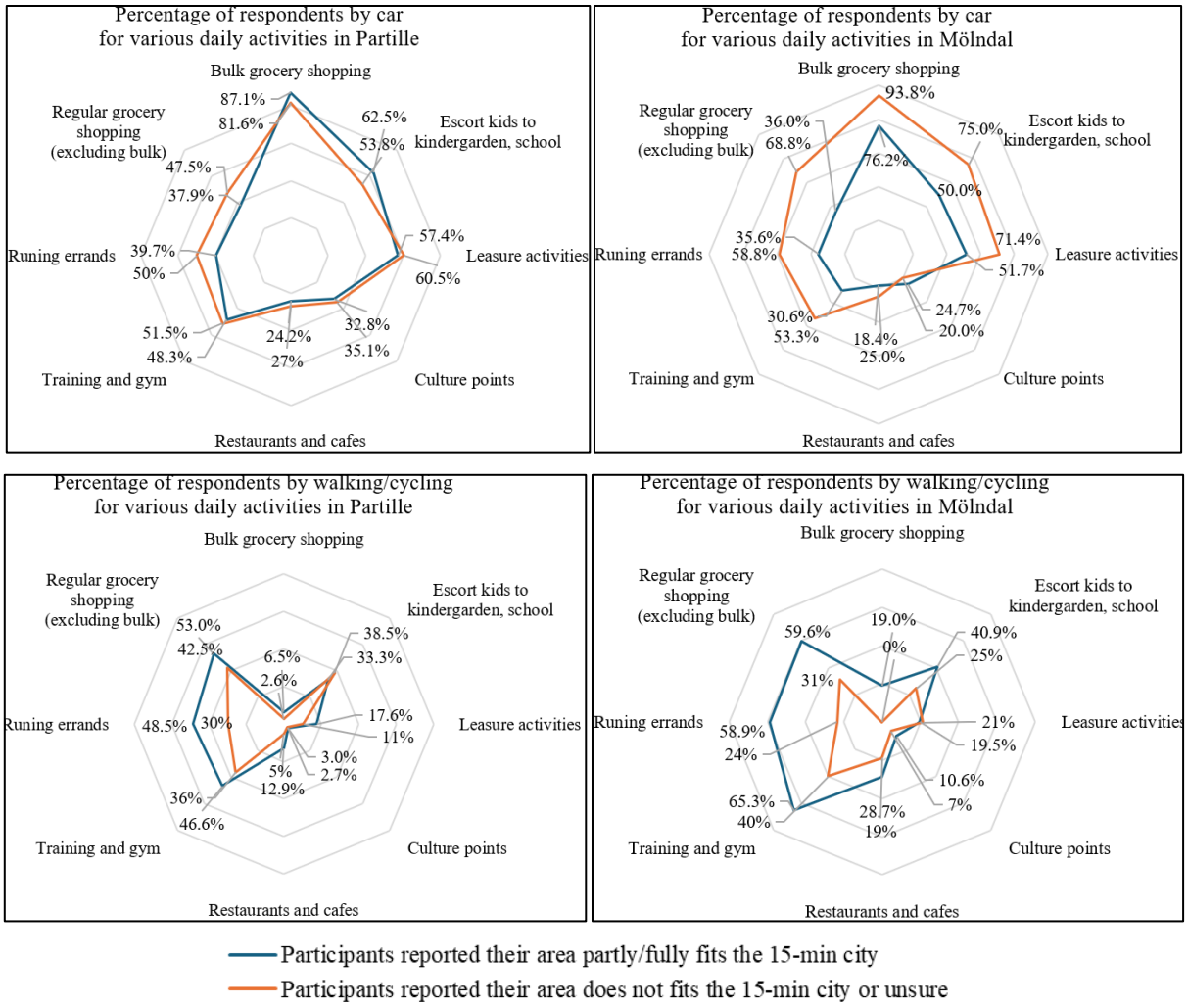


Figure 9. Percentage of respondents using car and walking/cycling for various daily activities in Mölndal and Partille, considering 15-Minute city perception. The category "Other/Not doing this activity" was excluded, as it combines alternative modes with non-participation in the activity.

For participants who reported that their neighbourhood is not aligned with the 15-minute city concept or who were unsure, a follow-up question was asked: *"If your local area were more like a 15-minute city, would it significantly affect your travel habits?"*. In the selected area in Mölndal, 17 out of 107 participants responded "no" or "unsure" regarding whether their neighbourhood aligns with the 15-minute city concept. Among these respondents, 11 participants (65%) indicated that they were unsure or believed that their travel habits would not be significantly affected even if their local area became more like a 15-minute city. Similarly, in the selected area in Partille, 40 out of 109 participants answered "no" or "unsure" when asked whether their neighborhood aligns with the 15-minute city concept. Of these respondents, 19 participants (approximately 50%) stated that a transition toward a 15-minute city would not significantly influence their travel habits.

Participants were asked about their personal experience with the 15-minute city through an open-ended question. The question was formulated as: "Do you think that the 15-minute city works well for you? Feel free to comment if you like." Responses to this question expressed a range of views regarding how well the 15-minute city concept works for them in practice. Many respondents reported that it works well, noting that most of their daily needs are met within their local area. Some even described their neighbourhoods as highly accessible, with convenient access to groceries, services, and public transport. Others supported the concept but expressed certain reservations. These included concerns about feeling confined to their local area, the need to travel to central Gothenburg for cultural and social

activities, and the challenges faced by families with children who often rely on cars for errands and activities. Additionally, some participants highlighted a continued dependence on cars, particularly when transporting goods or large items.

Participants were asked to reflect on shortcomings in their local area through an open-ended question, formulated as: “What do you think is missing for your neighbourhood to fully be a 15-minute city?” In response participants described several key elements they believe are missing in their neighbourhoods to fully meet the criteria of a 15-minute city. In terms of retail and services, they highlighted the need for larger, well-stocked grocery stores, more diverse and high-quality shops including pharmacies and specialty stores, and convenient local options like self-service stations for basic goods. Regarding food and social venues, participants expressed a desire for better cafés, restaurants, and pubs offering both variety and quality, as well as spaces for social interaction. In the area of culture and entertainment, they noted the absence of cultural venues such as theatres, cinemas, and event spaces, along with a lack of vibrant nightlife and local cultural activities. For education and personal development, participants mentioned the need for evening courses for adults (e.g., languages, crafts, yoga) and better facilities for sports and recreational activities. Public transport and infrastructure were also seen as insufficient, with calls for more frequent and better-connected bus lines, affordable public transport for short trips, improved cycling infrastructure including lighting and winter maintenance, and safer roads and pedestrian paths. From an urban planning and local economy perspective, participants expressed a desire for reduced reliance on central Gothenburg and suggested supporting small businesses through lower rents or subsidies, promoting mixed-use development with integrated shops and services, and revitalising former shopping streets and local centres. Finally, under community and vision, they emphasised the need for a more vibrant and engaging urban environment, better planning to avoid “sleeping neighbourhoods” composed only of housing, and a balanced approach that respects existing lifestyles—such as those who prefer detached homes with gardens.

4.3. Result of map analysis

Figure 10 and Figure 11 present the results of the map analysis for Mölndal and Partille, respectively. They illustrate walking and cycling times to the nearest location in any of the ten categories described in the Methods section (grocery stores, shopping areas, places to eat, pharmacies, health centres, daycare, schools, natural areas, cultural venues and spaces for social activities). The study area in Mölndal, covering all postal codes, was divided into 2,439 grid cells, while Partille was divided into 1,037 cells. This area includes only the chosen postal codes and not the area surrounding them.

Analysing the grid cells within the chosen postal codes in Mölndal it is shown that, shopping areas with a variety of stores are accessible within a 15-minute walk for 56% of grid cells, with a maximum walking time of 40 minutes. Pharmacies are reachable within 15 minutes for 58% of grid cells (maximum 38 minutes), while health centres are accessible for 88% (maximum 27 minutes). Nature and recreational areas are within walking distance for 100% of grid cells. Most other points of interest are accessible within 15 minutes for over 95% of grid cells. A combined overview of accessibility across all categories is presented in Figure 10, which shows the walking and cycling time to reach the nearest point to all categories. It is based on the distance to all categories and recalculated into speed, assuming a walking speed of 5 km /h and cycling speed of 15 km/h. The distance to each category is presented in *Figure S1* in Appendix C, together with the combined map with distances to all categories.

Within the chosen postal codes in Partille, shopping areas are accessible within a 15-minute walk for 56% of grid cells (maximum 33 minutes). Cultural and social venues are reachable within a 15-minute walk for 48% of grid cells (maximum 27 minutes), while pharmacies are accessible within a 15-minute walk for 91% of grid cells (maximum 18 minutes). The remaining categories are accessible for 99% of grid cells within the 15-minute walking threshold. Figure 11 presents the combined results for Partille. Map spatial analyses for each category in the selected areas in Mölndal is presented in *Figure S2* in Appendix C.

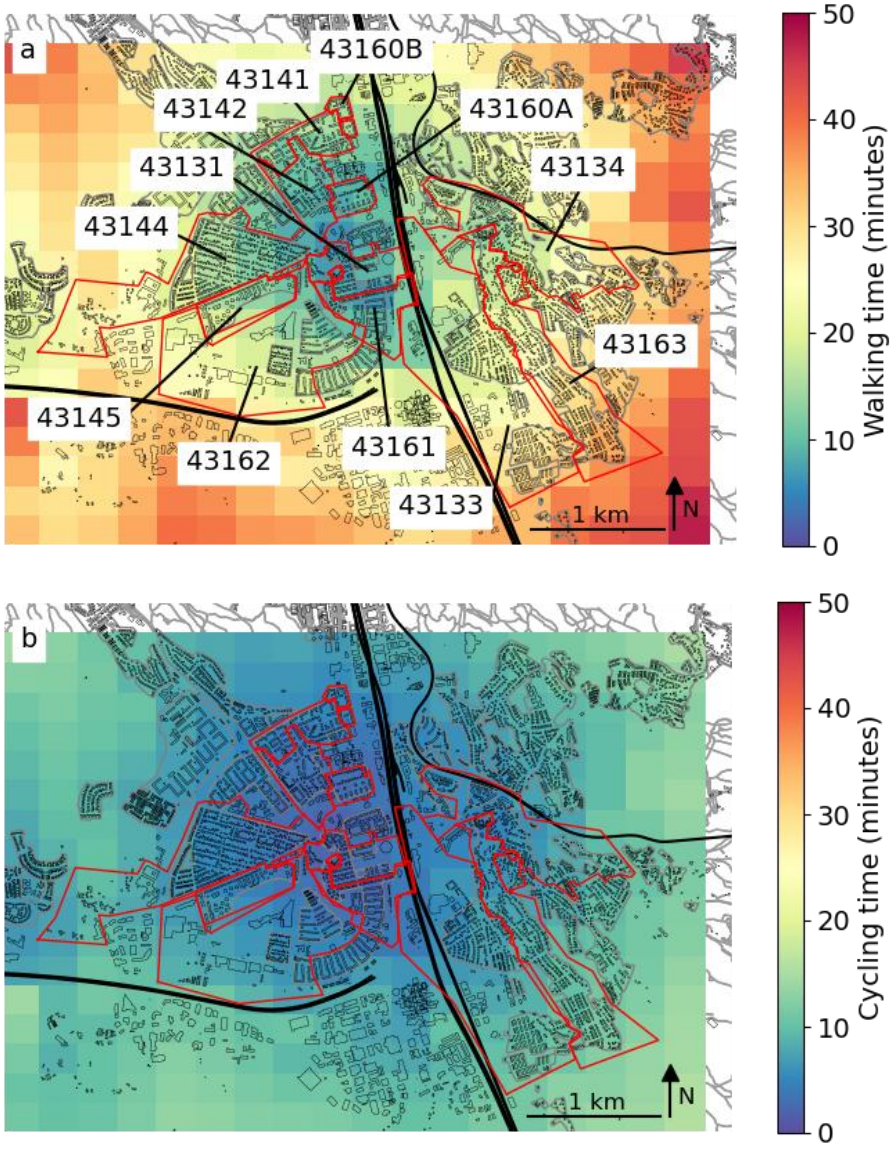


Figure 10. Map spatial analyses in the selected area in Mölndal. (a) Maximum walking time and (b) maximum cycling time to reach any of the ten specified categories Each marked zone represent a postal codes.

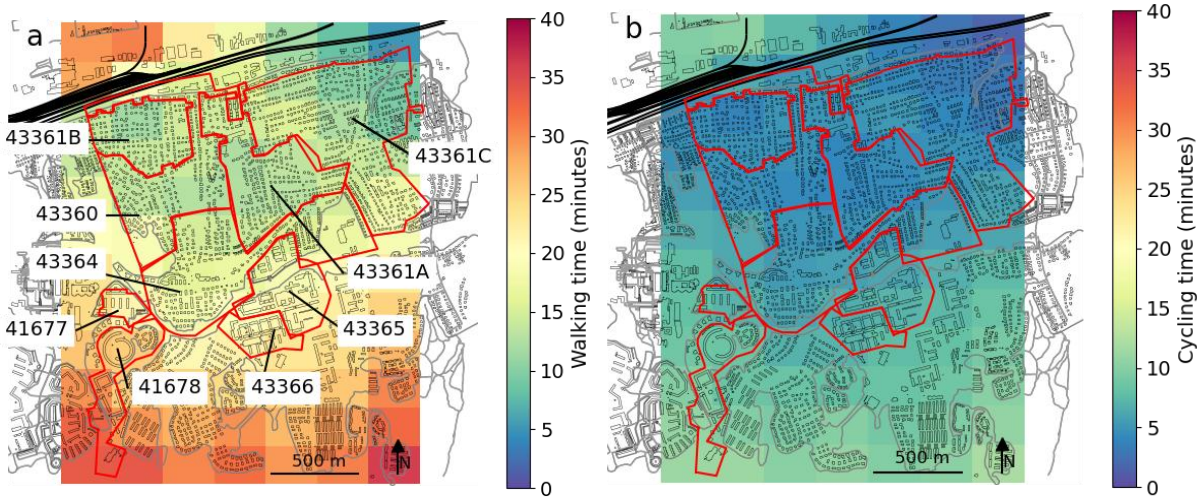


Figure 11. Map spatial analyses *in the selected area in Partille. (a) Maximum walking time and (b) maximum cycling time to reach any of the ten specified categories. Each marked zone represents a postal code.*

To better understand how proximity influences travel behaviour, in this project, survey data was compared with map-based accessibility analysis, using postal codes as the common reference point. Since participants provided their postal codes, it was possible to analyse responses separately for each area and compare them with the accessibility results presented in Figure 10 and Figure 11. This comparison provides valuable insights into how the 15-minute city concept manifests in practice.

According to the survey data (Question 8, see Appendix A), transport mode choices were reported for a set of predefined activities. In Mölndal, averaging across these activities, approximately 43% of responses indicate car use, 35% cycling or walking, and about 22% public transport. In Partille, the corresponding values are approximately 50% for car use, 25% for cycling or walking, and 25% for public transport (see Appendix B).

In the calculation of these averages, the category “Other/Not doing this activity” was excluded, as it combines alternative transport modes with non-participation in the activity, which may distort the interpretation of transport mode choice. Furthermore, these averages represent simple (unweighted) means across the selected activities and do not represent the share of all daily activities, as the survey did not capture the total number of activities or trips undertaken by respondents. Instead, they reflect the distribution of reported transport mode choices across the selected activities.

When survey responses are disaggregated by postal code and compared with the accessibility analysis, a clear pattern emerges. In Mölndal, in postal code areas with high accessibility (where nearly 100% of destinations are reachable within a 15-minute walk), approximately 45% of responses indicate walking or cycling, and approximately 35% indicate car use. In contrast, in areas located beyond the 15-minute walking threshold, approximately 25% of responses indicate walking or cycling, while about 55% indicate car use. Interestingly, the use of public transport remained relatively consistent across both high- and low-accessibility areas, at around 20% of responses.

In Partille, a different trend is observed. In postal code areas where the map-based analysis indicates that essential services are accessible within a 15-minute walk, approximately 20% of responses indicate walking or cycling, while about 60% indicate car use and about 20% public transport use. In contrast, in areas where walking to essential services takes more than 15 minutes, approximately 30% of responses indicate walking or cycling, 30% indicate car use, and about 45% indicate public transport use. This pattern differs from that observed in Mölndal and suggests that accessibility alone does not determine travel behaviour. The higher use of public transport in lower-accessibility areas may compensate for

longer distances, while the overall pattern may also reflect factors not captured by the accessibility analysis, such as urban structure, commuting patterns, sociodemographic characteristics, car availability, parking conditions, and differences among respondents across postal-code areas.

5. Discussion and policy implementation

Sweden is currently undergoing a significant urbanisation process, with approximately 85% of the population now living in urban areas (SCB, 2025b). This shift has increased the demand for more sustainable and accessible urban environments, making concepts such as the 15-minute city increasingly relevant.

The findings from this study showed that in the selected areas in both municipalities of Partille and Mölndal, all points of interest are accessible within 15 minutes by bicycle, and about 85% are accessible by foot, indicating that cycling significantly enhances local accessibility compared to walking alone. This finding are aligned with findings from Larsson et al. (2025) who reported that Partille municipality has accessibility of 100% and Mölndal municipality has accessibility of 95% by bike. The results also showed that people living outside walking distance, but within biking distance were less likely to perceive that they lived in a 15-minute city, showing that 15 minutes by bike may not be optimal for planning perspectives.

The most notable difference between the two municipalities lies in accessibility to cultural and social venues. In the selected areas in Mölndal, 98% of grid points are within walking distance of such amenities, compared to only 48% in the selected areas of Partille. This disparity is particularly noteworthy when viewed alongside the survey results in Figure 7, which show a higher proportion of Mölndal residents perceiving their municipality as meeting the criteria of a 15-minute city. These findings suggest that access to cultural amenities may be a significant factor influencing residents' perceptions of overall neighbourhood accessibility.

The 15-minute city is a relatively new concept, and most previous studies in this field have primarily focused on proximity, map-based accessibility, and spatial analysis. However, findings from this project, including insights highlighted during interviews, suggest that a more thorough approach is needed to fully understand how the concept works in practice. Future research should explore the relationship between proximity, local demand, and individual needs, and investigate why people continue to rely on private cars even when they live close to essential services. It is also important to establish a clear definition of what constitutes "essential needs" within the framework of the 15-minute city, and to examine which factors could effectively lead to a reduction in car use. Furthermore, it is worth questioning whether it is truly feasible to bring essential services closer to all residents in a way that meets diverse needs.

One major gap in previous studies is the lack of attention to commuting patterns, particularly how people travel to their workplaces. According to the survey results of this project, 40% of participants commute by car, 25% use public transport, 30% cycle and 5% walk to work. Among car users, the largest share of trips (62%) falls within the 15–30 minute travel time category. Public transport trips are most commonly reported in the 30–60 minute category, accounting for almost half (49%) of all public transport commutes. Cycling trips are distributed across multiple time categories, with a majority (41%) occurring in the 15-30 minutes category. Walking trips are predominantly reported in the 0–15 minute category, representing over 70% of all walking commutes. Moreover, the reported cycling times show a good match with the survey responses, where about 50% of participants indicated that a convenient cycling duration falls between 15 and 30 minutes. It is likely that owning a car, because it is needed for work, may also increase the use of a car for other activities. This suggests that integrating the 15-minute city concept with broader urban planning strategies, such as promoting remote work, advancing digitalisation, and improving delivery systems could help reduce the need for car travel. However, these approaches also have limitations: many occupations cannot be performed remotely, digital solutions may exclude certain population groups, and increased delivery activity can generate additional freight and service-related travel. Furthermore, it is important to consider how people will use the time and distance saved through these changes—what types of trips they might make instead, and whether those trips align with the goals of sustainable urban mobility.

It is essential for policymakers and urban planners to consider several practical and demographic factors when implementing the 15-minute city model. As the population in Sweden ages (SCB, 2025a), ensuring that active transport modes remain accessible and safe for older adults, children, and individuals with

disabilities becomes increasingly important. This includes year-round maintenance of pedestrian and cycling infrastructure, such as high-quality winter maintenance (Mirzanamadi et al., 2018). Additionally, planners must account for daily and temporal variations in activity patterns, particularly for individuals working non-standard hours or night shifts, whose needs may not align with typical service availability. The quality, openness, and inclusivity of essential services also play a critical role—proximity alone is not sufficient if services are not welcoming, affordable, or relevant to the local population (Willberg et al., 2023).

Understanding and support from the public and political groups are important for the success of the 15-minute city concept. One frequently raised concern is that such policies may be perceived as limiting personal freedom, for example by keeping residents in fixed areas or discouraging longer-distance travel. While some reports dismiss these concerns as conspiracy theories (Melnyk, 2023), public scepticism regarding perceived mobility constraints can still affect acceptance of the concept.

Some studies argue that the 15-minute city model could unintentionally make social and economic inequalities worse (Willberg et al., 2023). For instance, wealthier residents may be more likely to benefit from well-developed neighbourhoods, while lower-income groups remain concentrated in areas with fewer services and amenities. Proximity to services alone does not mean everyone benefits equally. Planning must therefore take into account how different social groups experience and use their surroundings in order to ensure that proximity translates into equitable outcomes.

6. Conclusion

This study combines objective measurements of proximity to key urban services with an exploration of residents' subjective perceptions of the "15-minute city" concept in two neighbourhoods on the outskirts of Gothenburg, Sweden. The research was conducted through a combination of seven in-depth interviews, questionnaire-based data collection, and spatial map analysis.

The interviews focused on two main questions: how transport efficiency is defined, and how it relates to the concept of the 15-minute city. Interviewees described transport efficiency as a holistic concept closely linked to urban planning goals, emphasising active mobility, improved accessibility, reduced resource use, and broader societal benefits and supporting public health. They also saw the 15-minute city as an inclusive planning model focused on proximity, walkability, and flexible use of public spaces. The interviewees noted that a large share of people in many Swedish cities live close to basic services. However, car use remains high despite this proximity. Interviewees also pointed out that implementing the 15-minute city can be difficult due to political, institutional, and infrastructure-related challenges.

The survey focused on participants' perceptions of the 15-minute city concept and its practical functionality. It aimed to understand which transport modes are used for various daily activities and the reasons behind those choices. Results showed that approximately 75% of participants felt their neighbourhoods fully or partially align with the principles of a 15-minute city. However, 82% still use their private cars for bulk grocery shopping (see Figure 5), and even if a well-stocked grocery store was located nearby, most would continue using their cars for large or weekly grocery shopping. This suggests that proximity alone is not sufficient to promote the 15-minute city concept. It also should be noted that shopping volume also plays an important role. Bulk purchases are typically associated with larger loads, which can limit the practicality of walking, cycling, or public transport, regardless of distance.

Another interesting finding was that 40% of participants commute to work by car, and among them, approximately 80% reported travel times exceeding 15 minutes. The most commonly cited reason for choosing the car was travel time (41%) and convenience (32%). Likely there is a relation between commuting to work, car ownership and mode choice after work-hours. Thus, excluding work-related commuting from the 15-minute city framework may lead to a misinterpretation of its effectiveness. To improve the functionality of the 15-minute city, it is important to consider alternatives such as increased opportunities for remote work and digitalisation. It should be noted that such options also have inherent limitations, as remote work is only feasible for certain types of occupations and cannot fully substitute for physical commuting.

The final part of this study involved a spatial map analysis to examine the proximity of basic services to individuals living in selected areas within the municipalities of Mölndal and Partille. It is important to note that this proximity analysis excluded workplace locations due to limitations in data access. As presented in Section 4.3, approximately half of the residents in the study area cannot reach shopping areas (containing a variety of stores, such as clothing and furniture stores) within a 15-minute walk although these areas are accessible within a 15-minute bike ride. However, the spatial analysis alone cannot determine whether residents would choose to cycle to these destinations or whether the types of goods purchased would influence their choice of transport mode. The size or weight of purchased items may discourage cycling and contribute to continued car use, but this possible explanation was not directly examined in the study.

The results of this study indicate that, although residents of the two selected areas live close to essential services such as restaurants, nature areas, grocery stores, and health centres, car dependency remains high—particularly for commuting to work and accessing larger shopping areas. Even when services are within walking or cycling distance, the convenience, short travelling time, and functionality of private cars continue to outweigh the benefits of proximity for many individuals. However, those with shorter travel times are more likely to walk or cycle and to perceive that they live in a 15-minute city, even though all locations to which the questionnaire was distributed fell within our definition of a 15-minute city. Thus, to further improve the understanding of 15-minute cities, planning strategies and research need to focus on

understanding individuals' behaviour, what factors other than proximity that can affect travel behaviour, how the city-scale and neighbourhood scale interact and how to make the planning inclusive for all social groups.

7. Declaration of generative AI

During the preparation of this work, the first author used “copilot.microsoft.com” only to improve the grammar and readability of the English text. After using this tool/service, the author reviewed and edited the content and takes full responsibility for the content of the published article.

References

Ali Sayigh. (2017). *Sustainable High Rise Buildings in Urban Zones, Advantages, Challenges, and Global Case Studies* (First). Springer Cham. https://doi.org/10.1007/978-3-319-17756-4

ArkDes. (2023). *Street Moves*. https://arkdes.se/en/projects/street-moves/

C40 Cities Climate Leadership Group, C. K. H. (2021). *Introducing Spotlight On: 15-minute cities*. https://www.c40knowledgehub.org/s/article/Introducing-Spotlight-On-15-minute-cities?language=en_US

Ferrer-ortiz, C., Marquet, O., Mojica, L., & Vich, G. (2022). Barcelona under the 15-Minute City Lens: Mapping the Accessibility and Proximity Potential Based on Pedestrian Travel Times. *Smart Cities*, 5(1), 146–161. https://doi.org/10.3390/smartcities5010010

Glazener, A., Wylie, J., Waas, W. Van, & Khreis, H. (2022). The Impacts of Car-Free Days and Events on the Environment and Human Health. *Current Environmental Health Reports*, 165–182. https://doi.org/10.1007/s40572-022-00342-y

Göteborg stad. (2024). *Kommunen i siffror*. https://goteborg.se/wps/portal/enhetssida/statistik-och-analys/statistik/kommunen-i-siffror

Hatzigeorgiou, A., Larsson, A., & Nagler, E. (2024). 15-minutersstaden – utopi eller verklighet? En geospatial analys för Sverige. In *Nationalekonomiska Föreningen*. https://www.nationalekonomi.se/artikel/15-minutersstaden-utopi-eller-verklighet-en-geospatial-analys-for-sverige/

Hosford, K., Beairsto, J., & Winters, M. (2022). Is the 15-minute city within reach? Evaluating walking and cycling accessibility to grocery stores in Vancouver. *Transportation Research Interdisciplinary Perspectives*, 14(December 2021), 100602. https://doi.org/10.1016/j.trip.2022.100602

InfraSweden. (2022). *Rikare Grannskap*. https://www.ri.se/sites/default/files/2022-11/Rikare grannskap slutrapport 2022-11-16.pdf

Knap, E., Baran, M., Geurs, K. T., Mulders, A., & Drift, S. Van Der. (2023). A composite X-minute city cycling accessibility metric and its role in assessing spatial and socioeconomic inequalities – A case study in Utrecht , the Netherlands. *Journal of Urban Mobility, 3*. https://doi.org/10.1016/j.urbmob.2022.100043

Larsson, A., Hatzigeorgiou, A., & Nagler, E. (2025). A geospatial analysis of Swedish 15-minute cities. *Discover Cities, 2*(1). https://doi.org/10.1007/s44327-025-00088-w

Lundin, P. (2008). *Bilsamhället: ideologi, expertis och regelskapande i efterkrigstidens Sverige*. Kungliga tekniska högskolan (KTH).

Melén, A., & Madsen, V. (2023). *15-minutersstaden i Göteborg*. University of Gothenburg.

Melnyk, K. (2023). *15-minute City: Conspiracy or Much Ado About Nothing?* https://www.drivesweden.net/nyheter/15-minute-city-conspiracy-or-much-ado-about-nothing

Ministry of the Environment and Energy. (2018). The Swedish Climate Policy Framework. *In Government Offices of Sweden*. https://www.government.se/495f60/contentassets/883ae8e123bc4e42aa8d59296ebe0478/the-swedish-climate-policy-framework.pdf

Mirzanamadi, R., & Gustafsson, M. (2022). *Users' experiences of tyre wear on electric vehicles - A survey and interview study*. https://vti.diva-portal.org/smash/get/diva2:1658879/FULLTEXT01.pdf

Mirzanamadi, R., Hagentoft, C. E., Johansson, P., & Johnsson, J. (2018). Anti-icing of road surfaces using Hydronic Heating Pavement with low temperature. *Cold Regions Science and Technology, 145,* 106–118. https://doi.org/10.1016/j.coldregions.2017.10.006

Mölndal kommun. (2025). *Befolkning statistik*. https://www.molndal.se/startsida/kommun-och-politik/fakta-om-molndal/statistik.html

Moreno, C., Allam, Z., Chabaud, D., Gall, C., & Pratlong, F. (2021). Introducing the "15-minute city": Sustainability, resilience and place identity in future post-pandemic cities. *Smart Cities, 4*(1), 93–111. https://doi.org/10.3390/smartcities4010006

Naturvårdsverket. (2023). *Sveriges utsläpp och upptag av växthusgaser*. https://www.naturvardsverket.se/data-och-statistik/klimat/sveriges-utslapp-och-upptag-av-vaxthusgaser/ #:~:text=Utsläppen från inrikes transporter minskar,2021 är utsläppsnivån närmast oförändrade.

Naturvårdsverket. (2025). *Inrikes transporter, utsläpp av växthusgaser*. https://www.naturvardsverket.se/data-och-statistik/klimat/vaxthusgaser-territoriella-utslapp-och-upptag/

Partille Kommun. (2025). *Befolkningsprognos 2024-2033*. https://www.partille.se/kommun--politik/kommunfakta/befolkning/

Patricio, J., Axelsson, L., Blomé, S., & Rosado, L. (2018). Enabling industrial symbiosis collaborations between SMEs from a regional perspective. *Journal of Cleaner Production, 202,* 1120–1130. https://doi.org/10.1016/j.jclepro.2018.07.230

Radics, M., Christidis, P., Alonso, B., & Dell'Olio, L. (2024). The X-Minute City : Analysing Accessibility to Essential Daily. *Land, 13*(10). https://www.mdpi.com/2073-445X/13/10/1656

Postaria, R. (2021). *15-minute city" – how do we get there*. https://www.citiesforum.org/news/15-minute-city/

Saiduddin, L. (2022). *The Post-Pandemic City – Covid-19 and its possible effects on urban planning in a Swedish context* [Swedish University of Agricultural Sciences]. https://stud.epsilon.slu.se/18537/1/saiduddin-l-20221215.pdf

SCB. (2023a). *Fordon enligt bilregistret efter fordonsslag, status och månad*. https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START__TK__TK1001__TK1001A/Fordon/chart/chartViewColumn/

SCB. (2023b). *Personbilar i trafik efter region, ägarkategori och år*. https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START__TK__TK1001__TK1001A/PersBilarA/chart/chartViewColumn/

SCB. (2025a). *Sweden's population in summary* 1960-2024. https://www.scb.se/en/finding-statistics/statistics-by-subject-area/population-and-living-conditions/population-composition-and-development/population-statistics/pong/tables-and-graphs/population-statistics---summary/swedens-population-in-summary-1960-202

SCB. (2025b). *Urbanisering – från land till stad*. https://www.scb.se/hitta-statistik/artiklar/2015/Urbanisering--fran-land-till-stad/

SOU. (2022). *Rätt för klimatet*. https://www.regeringen.se/rattsliga-dokument/statens-offentliga-utredningar/2022/05/sou-202221/

Staricco, L. (2022). 15- , 10- or 5-minute city ? A focus on accessibility to services in Turin , Italy. *Journal of Urban Mobility, 2*(April), 100030. https://doi.org/10.1016/j.urbmob.2022.100030

Statistics Sweden. (2025). *The Swedish Occupational Register with statistics*. https://www.scb.se/en/finding-statistics/statistics-by-subject-area/labour-market/labour-force-supply/the-swedish-occupational-register-with-statistics/

Stockholms Handelskammare. (2023). *Making Stockholm Closer- Assessing the 15-minute city concept to strengthen local urban life*. https://stockholmshandelskammare.se/rapporter/ny-rapport-making-stockholm-closer/

SWECO. (2025). *Närastaden – Sweco och 15-minutersstaden*. https://www.sweco.se/15-minutersstaden/

Toreld, L. (2023). *The 15-minute city - resilient urban planning or utopian vision* [Swedish University of Agricultural Sciences]. https://stud.epsilon.slu.se/18884/1/toreld-l-20230605.pdf

Trafikanalys. (2024). *Resvanor i Sverige 2024*. https://www.trafa.se/globalassets/statistik/resvanor/2025/resvanor-i-sverige-2024.pdf

Willberg, E., Fink, C., & Toivonen, T. (2023). The 15-minute city for all ? – Measuring individual and temporal variations in walking accessibility. *Journal of Transport Geography, 106(July 2022)*.

Yang, C. (2023). *Accelerating Sweden's climate action in transportation*. Stockholm Environment Institute. https://www.sei.org/perspectives/swedens-transport-climate-action/

Yannis, G., & Chaziris, A. (2022). Transport System and Infrastructure. *Transportation Research Procedia, 60*(2021), 6–11. https://doi.org/10.1016/j.trpro.2021.12.002

Appendix A. Questionnaire

No.	Questions	Alternatives
1	Are you currently working or studying?	1- Working
		2- Studying
		3- Other, for example retired or seeking employment
2	How often do you work from home on average? (Think of the past month)	1- I do not have the possibility to work from home
		2- I have the possibility, but I do not work from home that often
		3- 1 day per week
		4-2 days per week
		5- 3 days per week
		6-4 days per week
		7- 5 days per week or more
3	How long does the journey to your work/studies normally take in one direction?	1- 0-15 minutes
		2- 15-30 minutes
		3- 30-60 minutes
		4- More than 60 minutes
4	What mode of transportation do you usually use when traveling to and from your work/studies? (Consider the past month) If you use multiple modes of transportation, specify the one you spend the most time on	1-Car
		2- Public transportation
		3- Bicycle
		4- Walking
		5- Other, please specify
5	What is the reason you go by car to work/studies? (You can choose multiple options)	1- It is convenient
		2- I like it
		3- It is cheaper than other reasonable alternatives
		4- It is fast
		5- I need some of the car's features, such as cargo space, room for multiple people etc.
		6- It is my only possible alternative
		7- Other, please specify
6	What is the reason you use public transportation to get to work/studies? (You can choose multiple options)	1- It is convenient
		2- I like it
		3- It is cheaper than other reasonable alternatives
		4- It is fast
		5- I want to be environmentally friendly
		6- It is my only possible alternative
		7- Other, please specify
7	What is the reason you walk or bike to your work/studies? (You can choose multiple alternatives)	1- It is convenient
		2- I like it
		3- It is cheap
		4- It is fast
		5- I want to be environmentally friendly
		6- It is good for my health
		7- It is my only possible alternative
		8- Other, please specify
8	What mode of transportation do you usually use for these activities? (1- car, 2- public transport, 3- walking/biking, 4- other/ I do not do this)	1- Bulk shopping groceries (eg. buying for a whole week)
		2- Buying groceries (excluding bulk shopping)
		3- Runing errands (e.g. shop, collect packages, visit healthcare centre etc.)
		4- Training/physical activity (gym, sports centre, outdoor training etc.)
		5- Visit restaurants, cafés and other eateries
		6- Culture (cinema, theater, library etc.)
		7- Leisure activities
		8- Escort kids to kindergarden, school etc.

No.	Questions	Alternatives
9	How long do you think it is reasonable to cycle or walk to run an errand (e.g., shopping, picking up packages, picking up and dropping off children, etc.)?"	1- Maximum 5 minutes
		2- Maximum 15 minutes
		3- Maximum 30 minutes
		4- More than 30 minutes
10	Is there anything that makes you NOT want to cycle or walk in the local area? (You can choose multiple options)	1- No
		2- There is a lack of good and safe bike and walking paths.
		3- The area does not feel safe
		4- It is not convenient
		5- There is a lack of bike repair shops in the area
		6- Other, please specify
11	If there was a large grocery store in the local area, would you walk, cycle, or take public transport to do your big/weekly shopping?	1- No, I need the car
		2- Yes, if there was good public transport
		3- Yes, I would cycle or walk
		4- Unsure
12	What places or activities do you travel far to because they are lacking or of low quality in your local area? (You can choose multiple options)	1- Larger/more well-stocked grocery stores
		2- Better schools and preschools
		3- Better training facilities (gyms, sport centres, outdoor training etc.)
		4- More/better restaurants, cafés and eateries
		5- More/better social and cultural activities
		6- More/larger natural areas (parks, forests, beaches etc.)
		7- Other, please specify
		8- None of the above
13	There is a concept called the 15-minute city, where the idea is that all daily needs (e.g., grocery stores, services, culture, entertainment, and recreational activities) should be within a 15-minute walk or bike ride from all residents. It would also mean good cycling and walking paths and frequent public transportation. Do you think that your neighbourhood is a 15-minute city?	1- No
		2- Yes, to some extent
		3- Yes, definitely
		4- Unsure
14	If your local area were more like a 15-minute city, would it significantly affect your travel habits?	1- No
		2-Unsure
		3- Yes
15	What do you think is missing for your neighbourhood to become a 15-minute city?	
16	What do you think is missing for your neighbourhood to fully be a 15-minute city?	
17	Do you think that the 15-minute city works well for you? Feel free to comment if you like.	
18	What is your gender?	1- Female
		2- Male
		3- Prefer not to say
19	How old are you?	
20	What is your highest completed level of education?	1- No education
		2- Primary school
		3- High school
		4- Post-secondary education (other than college/university)
		5- College/university
		6- Doctoral studies
21	What type of housing do you have?	1- Apartment
		2- House/row house
		3- Other
22	How many people live in your household?	
23	How many children live in your household?	

No.	Questions	Alternatives
24	What is the total income for all household members per month before tax?	1- 0 - 30 000 kr
		2- 30 000 - 60 000 kr
		3- 60 000 - 90 000 kr
		4- 90 000 - 120 000 kr
		5- more than 120 000 kr
		6- Prefer not to say
25	What is you postal code?	

Appendix B. Comparison of transport mode use between perceived 15-minute city alignment

Area: Partille

Tabel A- Partially or fully yes (Group 1)- Partille								
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	**Escort kids to kindergarden, school**
Car	54	25	27	28	15	22	39	15
PT	4	6	8	3	39	43	17	1
Cycling/ Walking	4	35	33	27	8	2	12	8
Other/Not do this	7	3	1	11	7	2	1	45
Sum	69	69	69	69	69	69	69	69
Sum (excl. "Other" and "Not doing this")	62	66	68	58	62	67	68	24

Tabel B- No or unsure (Group 2)- Partille								
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	**Runing errands**	**Training and gym**	**Restaurants and cafes**	**Culture points**	**Leisure activities**	**Escort kids to kindergarden, school**
Car	31	19	20	17	10	13	23	7
PT	6	4	8	4	25	23	11	1
Cycling/ Walking	1	17	12	12	2	1	4	5
Other/Not do this	2	0	0	7	3	3	2	27
Sum	40	40	40	40	40	40	40	40
Sum (excl. "Other" and "Not doing this")	38	40	40	33	37	37	38	13

The difference reported for each row is calculated as the proportion of respondents in Group 1 associated with a given activity–transport mode combination minus the corresponding proportion in Group 2. Specifically, the proportion for each group is obtained by dividing the number of respondents selecting a given activity and transport mode by the total number of respondents in that group, excluding "Other" and "Not doing this".

For example, for bulk grocery shopping by car, 54 respondents in Group 1 and 31 respondents in Group 2 reported using this mode. The resulting difference is therefore calculated as 54/62−31/38, which corresponds to a difference of approximately 5.5 percentage points.

Tabel C- Difference between Group 1 and Group 2 (percentage points)-Partille									
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	**Escort kids to kindergarden, school**	Average
Car	5.5	-9.6	-10.3	-3.2	-2.8	-2.3	-3.2	8.7	-2.2
PT	-9.3	-0.9	-8.2	-6.9	-4.7	2.0	-3.9	-3.5	-4.4
Cycling/ Walking	3.8	10.5	18.5	10.2	7.5	0.3	7.1	-5.1	6.6

Total responses from participants considering both groups

Tabel D- Total participants- Partille								
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	**Escort kids to kindergarden, school**
Car	85	44	47	45	25	35	62	22
PT	10	10	16	7	64	66	28	2
Cycling/ Walking	5	52	45	39	10	3	16	13
Other/Not do this	9	3	1	18	10	5	3	72
Sum	109	109	109	109	109	109	109	109
Sum (excl. "Other" and "Not doing this")	100	106	108	91	99	104	106	37

Total responses from participants considering both groups, excluding the category of "Other" and "Not doing this"

Tabel E- Total participants (percentage points)- Partille									
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	**Runing errands**	**Training and gym**	**Restaurants and cafes**	**Culture points**	**Leisure activities**	**Escort kids to kindergarden, school**	**Average**
Car	85.0	41.5	43.5	49.5	25.3	33.7	58.5	59.5	49.5
PT	10.0	9.4	14.8	7.7	64.6	63.5	26.4	5.4	25.2
Cycling/ Walking	5.0	49.1	41.7	42.9	10.1	2.9	15.1	35.1	25.2

Area: Mölndal

Tabel F- Partially or fully yes (Group 1)- Mölndal								
	Bulk grocery shopping	Regular grocery shopping (excluding bulk)	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	Escort kids to kindergarden, school
Car	64	32	32	22	16	21	45	11
PT	4	4	5	3	46	55	25	2
Cycling/ Walking	16	53	53	47	25	9	17	9
Other/Not do this	6	1	0	18	3	5	3	68
Sum	90	90	90	90	90	90	90	90
Sum (excl. "Other" and "Not doing this")	84	89	90	72	87	85	87	22

Tabel G- No or unsure (Group 2)- Mölndal								
	Bulk grocery shopping	Regular grocery shopping (excluding bulk)	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	Escort kids to kindergarden, school
Car	15	11	10	8	4	3	10	3
PT	1	0	3	1	9	11	1	0
Cycling/ Walking	0	5	4	6	3	1	3	1
Other/Not do this	1	1	0	2	1	2	3	13
Sum	17	17	17	17	17	17	17	17
Sum (excl. "Other" and "Not doing this")	16	16	17	15	16	15	14	4

Tabel H- Difference between Group 1 and Group 2 (percentage points)-Mölndal									
	Bulk grocery shopping	Regular grocery shopping (excluding bulk)	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	Escort kids to kindergarden, school	Average
Car	-17.6	-32.8	-23.3	-22.8	-6.6	4.7	-19.7	-25.0	-17.9
PT	-1.5	4.5	-12.1	-2.5	-3.4	-8.6	21.6	9.1	0.9
Cycling/ Walking	19.0	28.3	35.4	25.3	10.0	3.9	-1.9	15.9	17.0

Tabel I- Total participants- Mölndal								
	Bulk grocery shopping	**Regular grocery shopping (excluding bulk)**	**Runing errands**	**Training and gym**	**Restaurants and cafes**	**Culture points**	**Leisure activities**	**Escort kids to kindergarden, school**
Car	79	43	42	30	20	24	55	14
PT	5	4	8	4	55	66	26	2
Cycling/ Walking	16	58	57	53	28	10	20	10
Other/Not do this	7	2	0	20	4	7	6	81
Sum	107	107	107	107	107	107	107	107
Sum (excl. "Other" and "Not doing this")	100	105	107	87	103	100	101	26

Total responses from participants considering both groups, excluding the category of "Other" and "Not doing this"

Tabel J- Total participants (percentage points)- Mölndal									
	Bulk grocery shopping	Regular grocery shopping (excluding bulk)	Runing errands	Training and gym	Restaurants and cafes	Culture points	Leisure activities	Escort kids to kindergarden, school	Average
Car	79.0	41.0	39.3	34.5	19.4	24.0	54.5	53.8	43.2
PT	5.0	3.8	7.5	4.6	53.4	66.0	25.7	7.7	21.7
Cycling/ Walking	16.0	55.2	53.3	60.9	27.2	10.0	19.8	38.5	35.1

Appendix C. Supplementary Figures

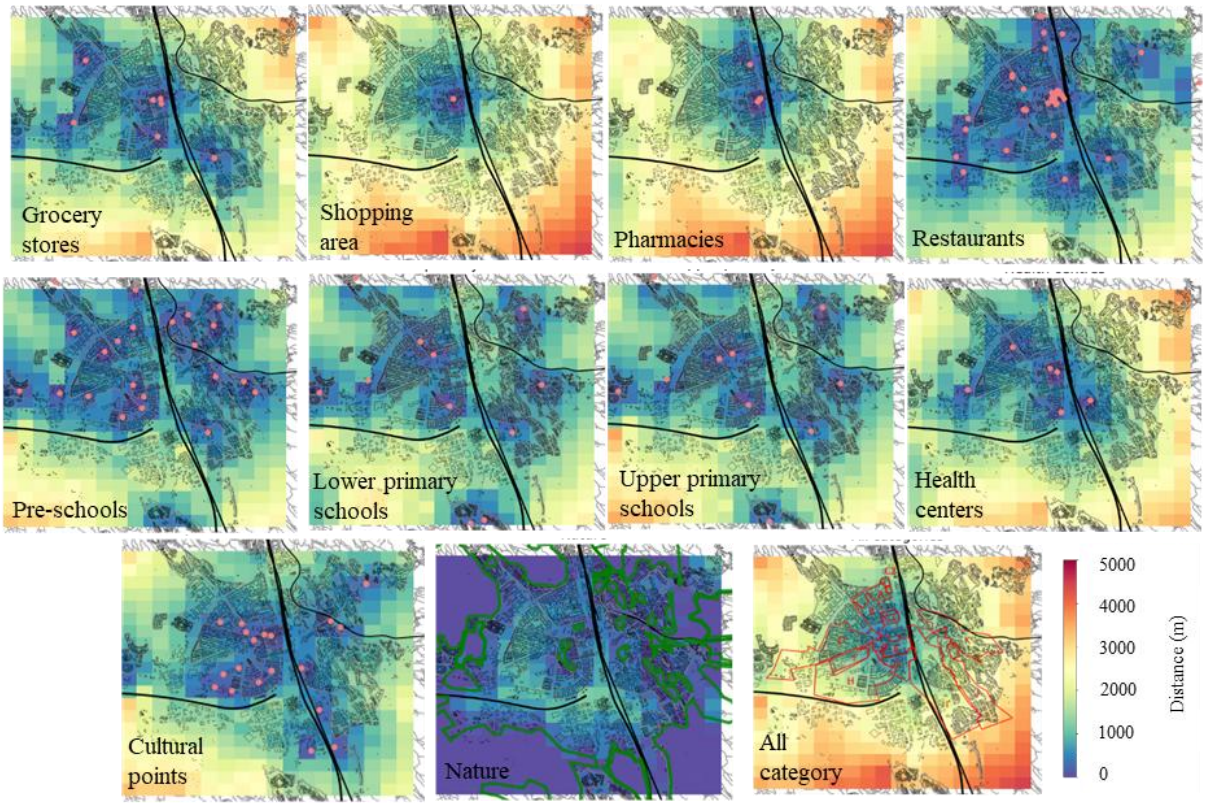


Figure S1. Map spatial analyses for each category in the selected areas in Mölndal.

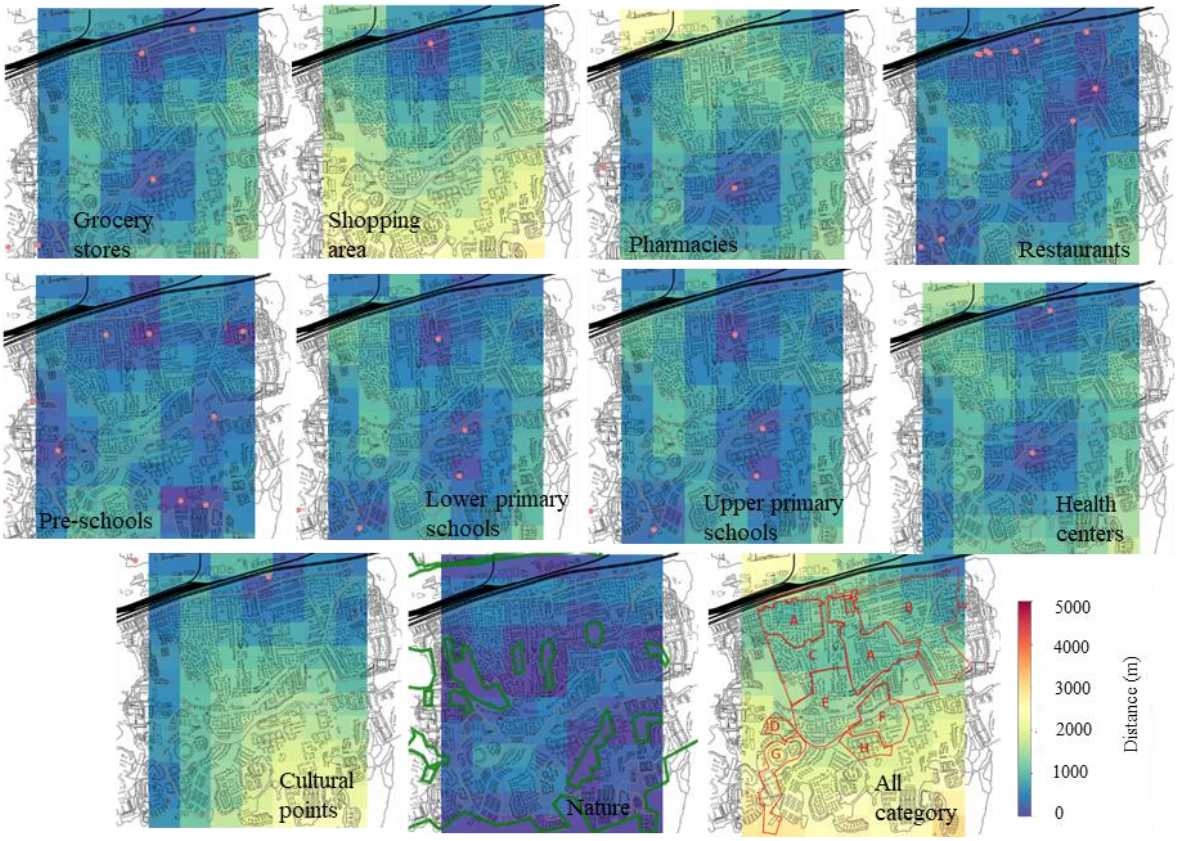


Figure S2. Map spatial analyses for each category in the selected areas in Partille.

VTI (Swedish National Road and Transport Research Institute) is Sweden's independent institute for applied transport research. On behalf of the Swedish Government, we conduct research and development that addresses real societal needs and contributes to an efficient, safe and sustainable transport system.

Our research covers all modes of transport and ranges from people, vehicles and infrastructure to planning, economics and policy instruments. Within a multidisciplinary organisation, we combine specialist expertise with a broad systems perspective and advanced research environments. Through a unique research infrastructure, we produce independent evidence and decision support used in both national and international transport policy.

VTI works in close collaboration with academia, public authorities and industry. We are around 250 people working together from offices in Linköping, Stockholm, Gothenburg, Lund and Skellefteå.

vti